Online Bilevel Optimization without Hyper-Gradients: Fully First-Order Single-Loop Methods

Abstract

We study online bilevel optimization (OBO), where at each round the upper-level objective is evaluated at the follower's implicit solution of a time-varying lower-level problem. We propose *VR–SF^{2}OBO*, a variance-reduced, fully first-order, single-loop algorithm that avoids linear-system solves and Hessian–vector products. Under a strongly convex lower level, we establish a sublinear *stochastic bilevel local-regret* bound that scales with the upper-level value drift, the follower path length, and gradient drift; no additional Hessian/Jacobian variation budgets appear in the final rate, and the bound recovers offline rates when drift vanishes. This contrasts with online hyper-gradient methods, which obtain sharper horizon dependence at the cost of stronger second-order oracles. Empirically, on online loss tuning for imbalanced data and meta-learning under distribution shift, our methods achieve accuracy comparable to hyper-gradient baselines while offering the lower per-round runtime than alternating-gradient and window-smoothed OBO competitors.

1 INTRODUCTION

Bilevel optimization (BO) aims to minimize an *upper-level* objective that is evaluated at the solution of a *lower-level* optimization problem. Originating in game theory [Von Stackelberg et al., 1953] and later formalized within mathematical optimization [Bracken and McGill, 1973], BO underpins a range of modern machine learning workflows: hyperparameter optimization and implicit differentiation [Lorraine et al., 2020, Franceschi et al., 2018], meta-learning [Ji et al., 2021], reinforcement learning [Stadie et al., 2020], and neural architecture search as well as loss-shaping pipelines [Liu et al., 2019, Li et al., 2021] all admit natural bilevel formulations. Classical theory largely consider the *offline* regime—objectives are fixed and the goal is to find a single stationary point (or minimizer) [Bracken and McGill, 1973, Maclaurin et al., 2015, Ji et al., 2021, Chen et al., 2025b]. By contrast, many real-world systems are inherently *online*: inputs and labels drift over time [Gama et al., 2014, Quiñonero-Candela et al., 2009], objectives evolve under variation budgets in online optimization [Besbes et al., 2015], reward structures change in nonstationary RL [Padakandla et al., 2019], and resource-aware deployments face hardware-dependent latency/energy constraints [Cai et al., 2020]. In these settings, the learner must update decisions on the fly from partial, noisy feedback, rather than repeatedly solving a fixed bilevel problem.

Online bilevel optimization (OBO) captures dynamic settings where objectives change over time, requiring the leader to continuously adjust the upper-level decision in response to the follower's (implicit) solution. As in online single-level optimization [Hazan, 2016], OBO proceeds via iterative decisions made without advance knowledge of outcomes [Ataee Tarzanagh et al., 2024b, Nazari et al., 2025]. We define online bilevel optimization (OBO) at round t by

$$\min_{x\in\mathbb{R}^{d_1}} \quad \phi_t(x) := f_t\big(x, y_t^*(x)\big) \tag{1}$$

$$\text{s.t.} \quad y_t^*(x) = \arg\min_{y\in\mathbb{R}^{d_2}} g_t(x,y). \tag{2}$$

Here, x and y are the upper- and lower-level variables, and f_t and g_t are the corresponding time-varying objectives for $t \in \{1,\dots,T\}$. Over T rounds, the leader selects $x_t \in \mathbb{R}^{d_1}$ without knowing $y_t^*(x_t)$; instead, it uses a follower-provided estimate y_t based on limited information about g_t, and incurs the outer loss $f_t(x_t, y_t)$.

This setting extends online single-level optimization to *nested* objectives and is correspondingly more challenging: to optimize $\phi_t(x)$, the leader only has access to an approximate follower state y_t and the history of past objectives $\{(f_k, g_k)\}_{k\le t-1}$. Leveraging this history can improve the tracking of the solution map $y_t^*(x)$ and thus refine y_t as its estimate, but residual tracking error $\|y_t - y_t^*(x_t)\|$ and temporal drift of (f_t, g_t) directly impact outer-level performance. Recent OBO methods typically alternate lower- and upper-level updates and evaluate performance via dynamic regret against a time-varying comparator. However, many of these approaches either assume strong oracle access—e.g., exact linear-system solves to compute hyper-gradients—or adopt *window-smoothed* (sliding-window) regret definitions that improve stability but dull responsiveness to rapid changes [Ataee Tarzanagh et al., 2024b, Lin et al., 2023b].

Challenges. OBO inherits the difficulties of online nonconvex learning and compounds them with the nested structure of BO: the *hyper-gradient* $\nabla\phi_t$ depends simultaneously on (i) the current follower solution $y_t^*(x)$ and (ii) second-order information of the follower objective g_t. Classic hyper-gradient estimators either use implicit differentiation with linear-system solves/Hessian–vector products [Pedregosa, 2016, Gould and Fernando, 2016, Lorraine et al., 2020] or differentiate through unrolled inner iterations [Maclaurin et al., 2015, Franceschi et al., 2017, 2018]. In online settings, methods such as OAGD and SOBOW add history/window smoothing and often perform multiple inner iterations per round to temper drift and variance [Ataee Tarzanagh et al., 2024a, Lin et al., 2023a, Bohne et al., 2024], which may not accurately reflect system performance when functions change rapidly. In contrast, the SOGD approach [Anonymous, 2025] reduces oracle requirements for hyper-gradient estimation and achieves sublinear regret *without* window smoothing. We propose **F^{2}OBO**, a *fully first-order, single-loop* OBO method that avoids Hessian–vector products and linear-system solves, together with its stochastic variance-reduced variant **VR–SF^{2}OBO**. Both algorithms adopt a penalty-based outer objective and take first-order outer steps using the same class of fully first-order hyper-gradient directions developed for offline BO [Chen et al., 2025b, Kwon et al., 2023, Chen et al., 2025a]. Analytically, we establish sublinear deterministic and stochastic *bilevel local regret* bounds for F^{2}OBO and VR–SF^{2}OBO, respectively.

Contributions.

- **Fully first-order, single-loop OBO.** We propose **F^{2}OBO**, which performs one penalized inner step, one tracker step, and one outer update per round using only gradients of (f_t, g_t)—no Hessian–vector products and no linear-system solves.
- **Deterministic local-regret with variation budgets.** Under standard assumptions (lower-level strong convexity; smoothness), we prove an average gradient local-regret bound that scales with outer value-variation and follower path-length, recovering the static scaling of batch bilevel methods when drift vanishes [Ji et al., 2021, Chen et al., 2025b].
- **Stochastic variance reduction.** We propose **VR–SF^{2}OBO**, a STORM-style variance-reduced variant with $\mathcal{O}(1)$-sized mini-batches, and establish a stochastic bilevel local-regret bound under standard oracle models.
- **Experiment results.** Compared to OAGD/SOBOW/OBBO, our methods avoid window smoothing and multi-iteration inner solves [Ataee Tarzanagh et al., 2024a, Lin et al., 2023a, Bohne et al., 2024], improving responsiveness and per-round efficiency in nonstationary regimes.

RELATED WORK

Foundations and offline bilevel optimization. Comprehensive treatments of BO and its links to MPECs appear in Dempe [2002], Colson et al. [2007]. Two dominant offline paradigms are implicit differentiation (ID/AID) [Pedregosa, 2016, Gould and Fernando, 2016, Lorraine et al., 2020] and iterative/unrolled differentiation (ITD) [Maclaurin et al., 2015, Franceschi et al., 2017, 2018]. Recent analyses provide convergence for nonconvex–strongly-convex bilevel problems and show that *fully first-order* algorithms can attain near-optimal complexity without Hessian-vector products [Ji et al., 2021, Chen et al., 2025b]. Variance-reduced bilevel frameworks and estimators have also been developed [Dagréou et al., 2022, Fang et al., 2018, Cutkosky and Orabona, 2019b].

Online bilevel optimization (OBO). OAGD introduces dynamic/static bilevel regret and path-length measures for both inner and outer minimizers [Ataee Tarzanagh et al., 2024a]. SOBOW proposes a single-loop, *window-averaged* hyper-gradient method that achieves sublinear bilevel local regret without requiring access to past objectives [Lin et al., 2023a], and subsequent work extends to Bregman geometries and stochastic variants with window averaging (OBBO/SOBBO)

[Bohne et al., 2024]. Our approach differs by (i) keeping one update per level without window smoothing, and (ii) adopting proxy-based outer steps with fully first-order directions [Chen et al., 2025b].

Online nonconvex learning and local regret. Because external regret is typically unattainable for general nonconvex losses, online analyses focus on gradient/prox-gradient stationarity notions [Hallak et al., 2021, Héliou et al., 2020, 2021]; we follow this line for OBO and use standard OCO tools for context [Hazan, 2016].

2 PROBLEM SETUP AND NOTATION

We consider an online bilevel problem over rounds $t = 1, \dots, T$. At each round t, we are given differentiable functions

$$f_t, g_t : \mathbb{R}^{d_1} \times \mathbb{R}^{d_2} \to \mathbb{R}.$$

For a leader decision $x \in \mathbb{R}^{d_1}$, the follower solves

$$y_t^*(x) \in \arg\min_{y \in \mathbb{R}^{d_2}} g_t(x, y),$$

and the (time-varying) *hyper-objective* is

$$\phi_t(x) \;:=\; f_t\big(x,\, y_t^*(x)\big).$$

We focus on the unconstrained leader case $X = \mathbb{R}^{d_1}$.[1] In this setting, the gradient local regret is defined as follows:

$$\mathsf{Reg}_T^{\text{loc}} \;:=\; \frac{1}{T} \sum_{t=1}^{T} \big\|\nabla \phi_t(x_t)\big\|^2. \tag{3}$$

Penalized proxy and notation. For a penalty parameter $\lambda > 0$, define the penalized lower-level objective and its value function

$$L_{\lambda,t}(x, y) := f_t(x, y) + \lambda\Big(g_t(x, y) - \min_z g_t(x, z)\Big), \tag{4}$$

$$L^*_{\lambda,t}(x) := \min_y L_{\lambda,t}(x, y), \tag{5}$$

and denote by $y^*_{\lambda,t}(x) \in \arg\min_y L_{\lambda,t}(x, y)$ a (measurable) minimizer. Note that for each fixed x, the term $\min_z g_t(x, z)$ is constant in y, so $y^*_{\lambda,t}(x)$ is equivalently defined by $y^*_{\lambda,t}(x) \in \arg\min_y \big(f_t(x, y) + \lambda g_t(x, y)\big)$. We will use the shorthands

$$\ell \;:=\; \max\{C_f,\, L_f,\, L_g,\, \rho_g\}, \qquad \kappa \;:=\; \ell/\mu.$$

Assumption 1 (Uniform regularity across t)**.** *Uniformly for all rounds t and all (x, y):*

(A1) ***Lower level strong convexity.*** *$g_t(x, \cdot)$ is μ–strongly convex: $\nabla^2_{yy} g_t(x, y) \succeq \mu I$.*

(A2) ***First–order smoothness.*** *∇f_t and ∇g_t are L_f– and L_g–Lipschitz in (x, y), respectively.*

(A3) ***Second–order regularity (in y).*** *$\nabla^2 g_t$ is ρ_g–Lipschitz in y: $\|\nabla^2 g_t(x, y) - \nabla^2 g_t(x, y')\| \le \rho_g \|y - y'\|$.*

(A4) ***Follower Lipschitz in value.*** *$f_t(x, \cdot)$ is C_f–Lipschitz in y: $|f_t(x, y) - f_t(x, y')| \le C_f \|y - y'\|$.*

Let $\ell := \max\{C_f, L_f, L_g, \rho_g\}$ and $\kappa := \ell/\mu$.

Variation budgets. Our analysis measures nonstationarity via the upper-level objective drifts and the path-length budgets:

$$V_T^f \;:=\; \sum_{t=2}^{T} \sup_{x,y} \big|\, f_t(x, y) - f_{t-1}(x, y) \,\big|, \tag{6}$$

$$H_{1,T} \;:=\; \sum_{t=2}^{T} \sup_{x} \big\|\, y_t^*(x) - y_{t-1}^*(x) \,\big\|. \tag{7}$$

$$H_{2,T} \;:=\; \sum_{t=2}^{T} \sup_{x} \big\|\, y_t^*(x) - y_{t-1}^*(x) \,\big\|^2. \tag{8}$$

[1] The constrained case can be handled by replacing the outer update with a projection onto X; we omit these standard details.

3 ALGORITHM: F^2OBO (WINDOW-FREE)

Variation budgets. Let $\{\phi_t\}_{t=1}^T$ denote the time-varying hyper-objectives $\phi_t(x) := f_t(x, y_t^\star(x))$ where $y_t^\star(x) \in \arg\min_y g_t(x, y)$. We quantify nonstationarity via two standard budgets in equation 6 and equation 7.

Performance metric (window-free local regret).

$$\mathsf{Reg}_T^{\mathrm{loc}} := \frac{1}{T}\sum_{t=1}^{T} \big\|\nabla \phi_t(x_t)\big\|^2.$$

Algorithm 1 F^2OBO (outer *unwindowed*; inner per-round)

1: **Inputs:** penalty $\lambda \geq 2L_f/\mu$; stepsizes $\{\eta_t^x, \eta_t^y, \eta_t^z\}$. Initialize (x_0, y_0, z_0).
2: **for** $t = 0, \ldots, T-1$ **do**
3: **Inner (penalized, per-round):** $y_{t+1} = y_t - \eta_t^y\big(\nabla_y f_t(x_t, y_t) + \lambda\, \nabla_y g_t(x_t, y_t)\big)$
4: **Inner (unpenalized, per-round):** $z_{t+1} = z_t - \eta_t^z\, \nabla_y g_t(x_t, z_t)$
5: **Outer (unwindowed):** $x_{t+1} = x_t - \eta_t^x\Big(\nabla_x f_t(x_t, y_t) + \lambda\big(\nabla_x g_t(x_t, y_t) - \nabla_x g_t(x_t, z_t)\big)\Big)$
6: **end for**

This is the single-step, time-varying specialization of F2BA; it uses the same fully first-order search direction as in Eq. (6) of Chen et al. [2025a].

4 ANALYSIS UNDER BOUNDED VARIATION BUDGETS (WINDOW-FREE)

Assumptions. We work under Assumption 1. Let L_ϕ be a uniform smoothness constant of $\{\phi_t\}_{t=1}^T$ (see Lemma C.9). We assume bounded variation budgets $V_T^f < \infty$, $H_{1,T} < \infty$, and $H_{2,T} < \infty$. We further take $\lambda \geq 2L_f/\mu$ and stepsizes satisfying, for all t,

$$\eta_t^x \leq \frac{1}{2L_\phi}, \qquad \eta_t^y \leq \frac{1}{L_f + \lambda L_g}, \qquad \eta_t^z \leq \frac{1}{L_g}.$$

Theorem 4.1 (Window-free local regret with bounded variation)**.** *Under the conditions above, Algorithm 1 yields*

$$\frac{1}{T}\sum_{t=1}^{T}\|\nabla\phi_t(x_t)\|^2 \;\leq\; \tilde{O}\Big(\frac{1}{\sqrt{T}}\Big) \;+\; \tilde{O}\Big(\frac{V_T^f + H_{1,T} + H_{2,T}}{\sqrt{T}}\Big) \;+\; O\Big(\frac{1}{\lambda^2}\Big) \;+\; O\Big(\frac{\|y_0 - y_{\lambda,1}^*(x_0)\|^2 + \|z_0 - y_1^*(x_0)\|^2}{T}\Big).$$

In particular, if $V_T^f = o(\sqrt{T})$ and $H_{1,T} + H_{2,T} = o(\sqrt{T})$, then $\mathsf{Reg}_T^{\mathrm{loc}} = o(1)$ as $T \to \infty$ (e.g., by taking $\lambda = \lambda_T \to \infty$ so that the $O(1/\lambda^2)$ bias vanishes).

Proof sketch. (i) *Proxy gradient accuracy.* By Lemma C.1 (proxy vs. true hyper-gradient), $\|\nabla L_{\lambda,t}^*(x_t) - \nabla\phi_t(x_t)\| \leq D_1/\lambda$. Moreover, by Lipschitzness of $\nabla_x f_t, \nabla_x g_t$ in y (Assumption 1), the algorithmic direction

$$d_t := \nabla_x f_t(x_t, y_t) + \lambda\big(\nabla_x g_t(x_t, y_t) - \nabla_x g_t(x_t, z_t)\big)$$

approximates $\nabla L_{\lambda,t}^*(x_t)$ up to the inner tracking errors $e_t^y := \|y_t - y_{\lambda,t}^*(x_t)\|$ and $e_t^z := \|z_t - y_t^*(x_t)\|$.

(ii) *Single-step descent.* Lemma C.9 bounds one outer step $\phi_t(x_{t+1}) - \phi_t(x_t)$ by (a) a descent term $-\eta_t^x\|\nabla\phi_t(x_t)\|^2$, (b) a proxy bias term $O(\eta_t^x/\lambda^2)$, and (c) inner tracking terms proportional to $\eta_t^x(e_t^y)^2$ and $\eta_t^x(e_t^z)^2$.

(iii) *Tracking error of inner iterates.* One penalized and one unpenalized step ensure $\|y_{t+1} - y_{\lambda,t}^*(x_t)\|^2$ and $\|z_{t+1} - y_t^*(x_t)\|^2$ contract up to the movement of $y_{\lambda,t}^*$ and y_t^* and the outer motion of x_t; see Lemma D.1. Summing the resulting recursion yields an $O(H_{2,T})$ contribution (plus initialization terms), and allows absorption of the $\sum_t \eta_t^x(e_t^y)^2$ and $\sum_t \eta_t^x(e_t^z)^2$ terms appearing in (ii).

(iv) *Nonstationarity control.* The change of objective across rounds contributes $\sum_{t=2}^T\big(\phi_t(x_t) - \phi_{t-1}(x_t)\big)$. By adding and subtracting $f_{t-1}(x_t, y_t^*(x_t))$ and using the C_f-Lipschitzness of $f_t(x,\cdot)$, this term is controlled by $V_T^f + C_f H_{1,T}$.

Choosing $\eta_t^x = \eta_x = \Theta(T^{-1/2})$ and combining (i)–(iv) yields the bound, together with the $O(1/\lambda^2)$ proxy bias and $O(1/T)$ initialization terms.

A complete proof (without $\tilde{O}(\cdot)$ shorthand) is provided in Appendix Section D.1. $\square$

Corollary 4.1 (Choosing λ and T for an ε-level bound)**.** *Fix $\varepsilon \in (0,1)$ and suppose bounds $V_T^f \le \bar{V}_T$, $H_{1,T} \le \bar{H}_{1,T}$, and $H_{2,T} \le \bar{H}_{2,T}$ are known. Pick $\eta_x = \Theta(T^{-1/2})$, and set*

$$\lambda \;\ge\; c_\lambda\, \varepsilon^{-1}, \qquad T \;\ge\; c_T\, \varepsilon^{-2} \left(1 + \bar{V}_T + \bar{H}_{1,T} + \bar{H}_{2,T}\right)^2,$$

for universal constants $c_\lambda, c_T > 0$ depending only on $(L_f, L_g, \mu, C_f, \rho_g)$. Then $\mathsf{Reg}_T^{\mathrm{loc}} \le \varepsilon$.

Remarks. (i) The result is *window-free*: no temporal smoothing in the outer step. (ii) The only nonvanishing bias term is $O(1/\lambda^2)$; it can be driven below ε by taking $\lambda = \Theta(\varepsilon^{-1})$. (iii) The budget terms quantify how quickly (f_t, g_t) and the follower solution map drift; the regret vanishes only when this drift is sufficiently small relative to T.

5 STOCHASTIC VR–SF2OBO

Stochastic oracles. For mini-batches B, write (e.g.) $\nabla_u h_t(w;B) = \frac{1}{|B|}\sum_{\xi\in B} \nabla_u h_t(w;\xi)$ with $h \in \{f,g\}$, $u \in \{x,y\}$, $w = (x,y)$. We assume unbiasedness and bounded second moments in the following assumption.

Assumption 2 (Stochastic oracle model)**.** *For any mini–batch B and any partial $u \in \{x,y\}$ of $h \in \{f,g\}$, there exists $\sigma_{h,u} > 0$ such that*

$$\mathbb{E}[\nabla_u h_t(w;B) \mid \mathcal{F}_t] = \nabla_u h_t(w), \qquad \mathbb{E}\left[\|\nabla_u h_t(w;B) - \nabla_u h_t(w)\|^2 \mid \mathcal{F}_t\right] \le \frac{\sigma_{h,u}^2}{|B|}.$$

The square-average smoothness is a standard hypothesis in variance-reduction analyses [Fang et al., 2018, Cutkosky and Orabona, 2019a] and is defined in the following.

Assumption 3 (Square-average smoothness (expected smoothness) of gradient estimators)**.** *For each round t and each partial $u \in \{x,y\}$ of $h \in \{f,g\}$ there exists $\overline{L}_{h,u} \ge 0$ such that for i.i.d. samples ξ and any two points $w = (x,y), w' = (x',y')$,*

$$\mathbb{E}_\xi\left[\|\nabla_u h_t(w;\xi) - \nabla_u h_t(w';\xi)\|^2\right] \;\le\; \overline{L}_{h,u}^2\, \|w - w'\|^2.$$

Consequently, for a mini-batch B of size $|B|$ reused inside a difference,

$$\mathbb{E}_B\left[\|\nabla_u h_t(w;B) - \nabla_u h_t(w';B)\|^2\right] \;\le\; \frac{\overline{L}_{h,u}^2}{|B|}\, \|w - w'\|^2.$$

Additional variation budgets. Our analysis in stochastic case, uses the value and path-length budgets defined in Equations (6) and (8) along with the following sequential difference between the individual gradients of the upper-level and lower-level functions:

$$D_{x,T}^f := \sum_{t=2}^T \sup_{x,y} \|\nabla_x f_t(x,y) - \nabla_x f_{t-1}(x,y)\|, \quad D_{y,T}^f := \sum_{t=2}^T \sup_{x,y} \|\nabla_y f_t(x,y) - \nabla_y f_{t-1}(x,y)\| \tag{9}$$

$$D_{x,T}^g := \sum_{t=2}^T \sup_{x,y} \|\nabla_x g_t(x,y) - \nabla_x g_{t-1}(x,y)\|, \quad D_{y,T}^g := \sum_{t=2}^T \sup_{x,y} \|\nabla_y g_t(x,y) - \nabla_y g_{t-1}(x,y)\|. \tag{10}$$

We also use the proxy (penalized) value-variation budget

$$V_T^{(\lambda)} \;:=\; \sum_{t=1}^T \sup_x \left| L_{\lambda_t,t}^*(x) - L_{\lambda_{t-1},t-1}^*(x) \right|, \tag{11}$$

Algorithm 2 Variance-Reduced SF^{2}OBO (VR–SF^{2}OBO)

Input: Horizon T; initial $(x_0, y_0, z_0, \lambda_1)$; step-size policies $\{\eta_t^x\}_{t=0}^{T-1}$, $\{\eta_t^y\}_{t=0}^{T-1}$, $\{\eta_t^z\}_{t=0}^{T-1}$; momentum/VR weights $\{\gamma_t^x\}_{t=0}^{T-1}$, $\{\gamma_t^y\}_{t=0}^{T-1}$, $\{\gamma_t^z\}_{t=0}^{T-1}$; batch-size policies for $B_t^{fx}, B_t^{fy}, B_t^{gx}, B_t^{gy}$; multiplier increments $\{\Delta_t^\lambda\}_{t=0}^{T-1}$.

Output: $(x_T, y_T, z_T, \lambda_T)$

Initialization: $d_0^z \leftarrow \nabla_y g_0(x_0, z_0; B_0^{gy})$; $\quad d_0^y \leftarrow \nabla_y f_0(x_0, y_0; B_0^{fy}) + \lambda_0 \nabla_y g_0(x_0, y_0; B_0^{gy})$;

$d_0^x \leftarrow \nabla_x f_0(x_0, y_0; B_0^{fx}) + \lambda_0(\nabla_x g_0(x_0, y_0; B_0^{gx}) - \nabla_x g_0(x_0, z_0; B_0^{gx}))$

for $t = 0$ **to** $T-1$ **do**

  Sample mini-batches $B_t^{fx}, B_t^{fy}, B_t^{gx}, B_t^{gy}$.

  $z_{t+1} \leftarrow z_t - \eta_t^z d_t^z$; $\quad y_{t+1} \leftarrow y_t - \eta_t^y d_t^y$; $\quad x_{t+1} \leftarrow x_t - \eta_t^x d_t^x$; $\quad \lambda_{t+1} \leftarrow \lambda_t + \Delta_t^\lambda$

  [d_t^z, d_t^y, and d_t^x are defined in Equation (12)]

end for

which captures the inter-round changes of the proxy objectives (including time-varying λ_t). Let $\gamma_t^x, \gamma_t^y, \gamma_t^z \in (0,1]$ be time-varying weights, and let $B_t^{fx}, B_t^{fy}, B_t^{gx}, B_t^{gy}$ denote mini-batches drawn at time t (reused *within* differences). Similar to the STORM updates in [Cutkosky and Orabona, 2019a], we design the following gradient updates:

$$d_t^z = \nabla_y g_t(x_t, z_t; B_t^{gy}) + (1-\gamma_t^z)\big(d_{t-1}^z - \nabla_y g_t(x_{t-1}, z_{t-1}; B_t^{gy})\big)\,, \tag{12a}$$

$$\begin{aligned} d_t^y = &\big[\nabla_y f_t(x_t, y_t; B_t^{fy}) + \lambda_t \nabla_y g_t(x_t, y_t; B_t^{gy})\big] \\ &+ (1-\gamma_t^y)\Big(d_{t-1}^y - \big[\nabla_y f_t(x_{t-1}, y_{t-1}; B_t^{fy}) + \lambda_{t-1} \nabla_y g_t(x_{t-1}, y_{t-1}; B_t^{gy})\big]\Big)\,, \end{aligned} \tag{12b}$$

$$\begin{aligned} d_t^x = &\big[\nabla_x f_t(x_t, y_t; B_t^{fx}) + \lambda_t\big(\nabla_x g_t(x_t, y_t; B_t^{gx}) - \nabla_x g_t(x_t, z_t; B_t^{gx})\big)\big] \\ &+ (1-\gamma_t^x)\Big(d_{t-1}^x - \big[\nabla_x f_t(x_{t-1}, y_{t-1}; B_t^{fx}) + \lambda_{t-1}\big(\nabla_x g_t(x_{t-1}, y_{t-1}; B_t^{gx}) - \nabla_x g_t(x_{t-1}, z_{t-1}; B_t^{gx})\big)\big]\Big)\,. \end{aligned} \tag{12c}$$

Theorem 5.1 (VR–SF^{2}OBO stochastic local regret)**.** *Assume Assumption 1 and the stochastic oracle conditions in Assumptions 2–3. Let $\{(x_t, y_t, z_t, \lambda_t)\}_{t=0}^T$ be generated by VR–SF^{2}OBO (Algorithm 2) with control variates defined in Equation* (12)*. Define $S_x := \sum_{t=0}^{T-1} \eta_t^x$ and the stochastic bilevel local regret*

$$\mathcal{R}_T^{\mathrm{loc}} := \frac{1}{S_x} \sum_{t=0}^{T-1} \eta_t^x\, \mathbb{E}\|\nabla \phi_t(x_t)\|^2.$$

If the step sizes satisfy $\eta_t^x \le \frac{1}{2L}$ (where L is a smoothness constant of $L_{\lambda_t,t}^*$), $\eta_t^y \le (L_f + \lambda_t L_g)^{-1}$, $\eta_t^z \le L_g^{-1}$, and the coupling conditions equation 92 hold, then

$$\begin{aligned} \mathcal{R}_T^{\mathrm{loc}} \le\ & \frac{C}{S_x}\Big(L_{\lambda_0,0}^*(x_0) - \inf_x L_{\lambda_T,T}^*(x) + V_T^{(\lambda)} + \mathbb{E}[\delta_0] + \sum_{t=0}^{T-1}\big(\Gamma_t^{(\lambda)} + \Gamma_t\big)\Big) \\ &+ \frac{CD_1^2}{S_x}\sum_{t=0}^{T-1}\frac{\eta_t^x}{\lambda_t^2} + \frac{C}{S_x}\sum_{t=0}^{T-1}\eta_t^x\,\mathbb{E}\|\epsilon_t^x\|^2 + \frac{C}{S_x}\sum_{t=0}^{T-1}\Big(\frac{\eta_t^y}{\lambda_t}\,\mathbb{E}\|\epsilon_t^y\|^2 + \eta_t^z\,\mathbb{E}\|\epsilon_t^z\|^2\Big)\,, \end{aligned} \tag{13}$$

where $\delta_0 := \|y_0 - y_{\lambda_0,0}^*(x_0)\|^2 + \|z_0 - y_0^*(x_0)\|^2$, and $\epsilon_t^u := d_t^u - G_t^u$ for $u \in \{x,y,z\}$ are the control-variate errors. Moreover, the last three sums are upper bounded in terms of the gradient-drift budgets in Equations (9) *and* (10) *and the oracle variances via Lemma E.2 and Lemma E.3.*

The complete proof is given in Appendix Section E.1.

Remark 5.1 (Budgets vs derivative drifts)**.** *The deterministic analysis can be expressed in terms of value/path-length budgets $(V_T^f, H_{1,T}, H_{2,T})$. In the stochastic variance-reduced setting, the recursion for the control variates naturally introduces* gradient-drift *budgets such as $D_{x,T}^f$ and $D_{x,T}^g$ in Equations* (9) *and* (10)*, which quantify inter-round changes of gradients.*

Remark 5.2 (Comparison to offline regime)**.** *When $f_t \equiv f$ and $g_t \equiv g$ (no drift), the budget terms vanish and the regret bound reduces to the corresponding offline fully first-order rate under the same stochastic oracle model.*

6 EMPIRICAL EVALUATIONS

In this section, we examine two experiments in online bilevel optimization and present their results: *Online Parametric Loss Tuning for Imbalanced Data* and *Online Meta-Learning*.

6.1 ONLINE PARAMETRIC LOSS TUNING FOR IMBALANCED DATA

Imbalanced datasets are ubiquitous in modern machine learning and introduce challenges for both generalization and fairness due to underrepresented classes and sensitive attributes. Deep neural networks are particularly prone to overfitting: they can appear fair and accurate during training yet exhibit poor performance at test time. A common remedy is to design a *parametric* loss that jointly balances accuracy and fairness while also mitigating overfitting. The AutoBalance approach [Li et al., 2021] was the first to automate this idea by introducing a parametric training loss that balances accuracy and fairness and reduces overfitting. In this empirical evaluation, we compare the stochastic variance-reduced version of our algorithm Algorithm 2 against several bilevel-optimization baselines in an online setting. The baselines include AutoBalance [Li et al., 2021], SOGD [Anonymous, 2025], SOBOW [Lin et al., 2023c], and OAGD (a variant of AutoBalance) [Tarzanagh et al., 2024]. SOBOW is a dynamic method that leverages conjugate gradients (CG), whereas OAGD is a static method that estimates hyper-gradients via a Neumann-series approximation. We consider an online bilevel optimization problem given in Equation (1), with a slight modification: rather than changing the leader and follower loss functions at each time step, we vary the datasets to which they are applied so the environment remains online. At each time step $t \in [T]$, we receive a new sample $(\mathbf{a}_t, b_t)$ drawn from $\mathcal{D}_t := \{\mathcal{D}_t^{\mathrm{val}}, \mathcal{D}_t^{\mathrm{tr}}\}$, where $\mathbf{a}_t \in \mathbb{R}^{d_2}$ is a feature vector and $b_t \in \{1, \ldots, J\}$ is the corresponding label.

Follower (training) loss. For such a sample, the follower employs a *parametric* cross-entropy (CE) loss:

$$g(x_t, y_t; \mathcal{D}_t^{\mathrm{tr}}) = -\log \frac{\exp\big(\gamma_{b_t} [y_t(\mathbf{a}_t)]_{b_t} + \Delta_{b_t}\big)}{\sum_{j=1}^{J} \exp\big(\gamma_j [y_t(\mathbf{a}_t)]_j + \Delta_j\big)}. \tag{9a}$$

Leader (validation) loss. At the outer level, the leader optimizes a balanced CE loss:

$$f\big(y_t(x_t); \mathcal{D}_t^{\mathrm{val}}\big) = -u_{b_t} \log \frac{\exp\big([y_t(\mathbf{a}_t)]_{b_t}\big)}{\sum_{j=1}^{J} \exp\big([y_t(\mathbf{a}_t)]_j\big)}. \tag{9b}$$

Here $x_t := (\Delta, \gamma)_{j=1}^{J}$ denotes per-class logit adjustments, and u_j is the reciprocal of the proportion of class j in the data [Li et al., 2021]. The model $y_t(x_t)$ is a 4-layer CNN, yielding a non-convex bilevel objective.

Experiments are conducted on MNIST [LeCun and Cortes, 2010] with mini-batch size 64. We compare cumulative runtime, balanced test accuracy, and balanced train accuracy, where balanced accuracy is defined as $\frac{1}{J}\sum_{j=1}^{J} \mathbb{P}_{\mathbf{a}_t \sim \mathcal{D}_j}\big[\arg\max_i [y_t(\mathbf{a}_t)]_i = j\big]$, with $\mathcal{D}_j$ the class-conditional sample distribution for class j and $\mathbb{P}[A]$ the probability of event A [Li et al., 2021]. To ensure consistency, we set hyperparameters adaptively and tune them accordingly. Training proceeds for 400 time steps in four phases of 100 steps each. Across phases, we anneal the class distribution from a highly imbalanced setting proportional to $\big(0.4^i\big)$ toward a more balanced setting proportional to $\big(0.7^i\big)$ for classes $(i = 0, \ldots, 9)$. Empirically, our approach achieves strong balanced accuracy while requiring substantially less runtime than all competing methods. This runtime advantage becomes even more pronounced when scaling to larger models and performing hyperparameter tuning, while the accuracy and fairness performance remains competitive. Tables 1 and 2 report, for each algorithm, the balanced train and test accuracies (mean±std across five runs) at iterations 100, 200, 300, and 400.

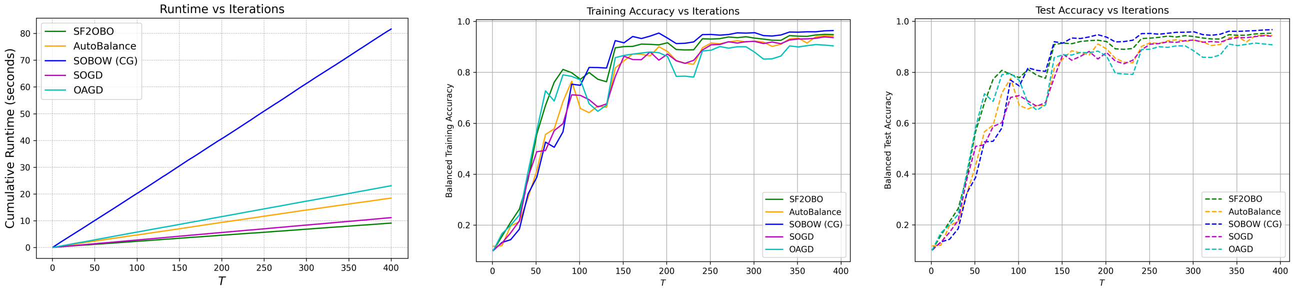

Figure 1: Performance (mean±std) on online parametric loss tuning with distribution shift on MNIST across five runs, comparing OAGD [Tarzanagh et al., 2024], SOBOW [Lin et al., 2023c], SOGD [Anonymous, 2025], AutoBalance [Li et al., 2021] with our stochastic variance-reduced method VR–SF^{2}OBO.

Algorithm\Iteration	**100**	**200**	**300**	**400**
SOBOW (CG)	0.847 ± 0.063	0.947 ± 0.023	0.958 ± 0.005	0.964 ± 0.005
VR–SF^{2}OBO	$\mathbf{0.845 \pm 0.048}$	$\mathbf{0.921 \pm 0.023}$	$\mathbf{0.940 \pm 0.004}$	$\mathbf{0.948 \pm 0.006}$
AutoBalance	0.760 ± 0.048	0.891 ± 0.005	0.920 ± 0.027	0.946 ± 0.007
SOGD	0.764 ± 0.044	0.896 ± 0.010	0.918 ± 0.016	0.931 ± 0.009
OAGD	0.825 ± 0.054	0.890 ± 0.018	0.903 ± 0.012	0.906 ± 0.008

Table 1: Balanced **train** accuracy (mean $\pm$ std) at iterations 100, 200, 300, and 400 for each algorithm in the *Online Parametric Loss Tuning for Imbalanced Data* experiment.

Algorithm\Iteration	**100**	**200**	**300**	**400**
SOBOW (CG)	0.874 ± 0.050	0.945 ± 0.009	0.960 ± 0.007	0.967 ± 0.003
VR–SF^{2}OBO	$\mathbf{0.849 \pm 0.039}$	$\mathbf{0.927 \pm 0.011}$	$\mathbf{0.945 \pm 0.004}$	$\mathbf{0.952 \pm 0.006}$
AutoBalance	0.789 ± 0.048	0.904 ± 0.008	0.921 ± 0.029	0.947 ± 0.002
SOGD	0.779 ± 0.038	0.896 ± 0.011	0.918 ± 0.014	0.937 ± 0.008
OAGD	0.820 ± 0.055	0.890 ± 0.003	0.909 ± 0.005	0.913 ± 0.008

Table 2: Balanced **test** accuracy (mean $\pm$ std) at iterations 100, 200, 300, and 400 for each algorithm in the *Online Parametric Loss Tuning for Imbalanced Data* experiment.

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CHECKLIST

The checklist follows the references. For each question, choose your answer from the three possible options: Yes, No, Not Applicable. You are encouraged to include a justification to your answer, either by referencing the appropriate section of your paper or providing a brief inline description (1-2 sentences). Please do not modify the questions. Note that the Checklist section does not count towards the page limit. Not including the checklist in the first submission won't result in desk rejection, although in such case we will ask you to upload it during the author response period and include it in camera ready (if accepted).

1. For all models and algorithms presented, check if you include:
   (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes/No/Not Applicable]
   (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes/No/Not Applicable]
   (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes/No/Not Applicable]
2. For any theoretical claim, check if you include:
   (a) Statements of the full set of assumptions of all theoretical results. [Yes/No/Not Applicable]
   (b) Complete proofs of all theoretical results. [Yes/No/Not Applicable]
   (c) Clear explanations of any assumptions. [Yes/No/Not Applicable]
3. For all figures and tables that present empirical results, check if you include:
   (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes/No/Not Applicable]
   (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes/No/Not Applicable]
   (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes/No/Not Applicable]
   (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes/No/Not Applicable]
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   (d) Information about consent from data providers/curators. [Yes/No/Not Applicable]
   (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Yes/No/Not Applicable]
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
   (a) The full text of instructions given to participants and screenshots. [Yes/No/Not Applicable]
   (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Yes/No/Not Applicable]
   (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Yes/No/Not Applicable]

A APPENDIX

B NOTATION AND ASSUMPTIONS (GLOSSARY)

C PROOFS FOR OFFLINE INGREDIENTS

Let $y^*_{\lambda,t}(x) \in \arg\min_y f_t(x,y) + \lambda\big(g_t(x,y) - g^*_t(x)\big)$. The offline paper shows [Chen et al., 2025a, Lemma 4.1]:

Lemma C.1. *Under Assumption 1 and $\lambda \geq 2L_f/\mu$, there exist constants $D_1, L_x = O(\ell\kappa^3)$ such that, for all x and each t,*

$$\left\|\nabla L^*_{\lambda,t}(x) - \nabla\phi_t(x)\right\| \;\leq\; \frac{D_1}{\lambda}, \qquad \nabla L^*_{\lambda,t} \text{ is } L_x\text{-Lipschitz in } x.$$

Moreover,

$$\nabla L^*_{\lambda,t}(x) = \nabla_x f_t\big(x, y^*_{\lambda,t}(x)\big) + \lambda\big(\nabla_x g_t\big(x, y^*_{\lambda,t}(x)\big) - \nabla_x g_t\big(x, y^*_t(x)\big)\big)\,.$$

Finally, $y^*_t(x)$ and $y^*_{\lambda,t}(x)$ are Lipschitz in x with constants L_g/μ and $4L_g/\mu$, respectively (Appendix B in Chen et al. [2025a]).

Lemma C.2. *Under Assumption 1, for $\lambda \geq 2L_f/\mu$, the function*

$$y \;\longmapsto\; L_{\lambda,t}(x,y) \;:=\; f_t(x,y) + \lambda\, g_t(x,y)$$

is $(\lambda\mu/2)$-strongly convex (in y).

Proof. Fix x. By Assumption 1, $g_t(x,\cdot)$ is μ-strongly convex, hence $\nabla^2_{yy} g_t(x,y) \succeq \mu I$ for all y. By Assumption 1(A2), ∇f_t is L_f-Lipschitz in (x,y) and hence in y; since f_t is twice continuously differentiable, we have $\|\nabla^2_{yy} f_t(x,y)\| \leq L_f$ for all y, which implies $\nabla^2_{yy} f_t(x,y) \succeq -\,L_f I$.

Therefore, for any y and any $v \in \mathbb{R}^{d_y}$,

$$v^\top \nabla^2_{yy} L_{\lambda,t}(x,y)\, v = v^\top\Big(\nabla^2_{yy} f_t(x,y) + \lambda\, \nabla^2_{yy} g_t(x,y)\Big)v \;\geq\; (-L_f + \lambda\mu)\,\|v\|^2.$$

Equivalently, $\nabla^2_{yy} L_{\lambda,t}(x,y) \succeq (\lambda\mu - L_f)I$ for all y. If $\lambda \geq 2L_f/\mu$, then $\lambda\mu - L_f \geq \frac{1}{2}\lambda\mu$, which shows that $L_{\lambda,t}(x,\cdot)$ is $(\lambda\mu/2)$-strongly convex. □

Lemma C.3. *Under Assumption 1, for $\lambda \geq 2L_f/\mu$, it holds that*

$$\left\|y^*_{\lambda,t}(x) - y^*_t(x)\right\| \;\leq\; \frac{C_f}{\lambda\,\mu}.$$

Proof. By the first-order optimality condition, we have

$$\nabla_y f_t\big(x, y^*_{\lambda,t}(x)\big) + \lambda \nabla_y g_t\big(x, y^*_{\lambda,t}(x)\big) \;=\; 0.$$

Therefore,

$$\left\|y^*_{\lambda,t}(x) - y^*_t(x)\right\| \;\leq\; \frac{1}{\mu}\left\|\nabla_y g_t\big(x, y^*_{\lambda,t}(x)\big)\right\| \;=\; \frac{1}{\lambda\,\mu}\left\|\nabla_y f_t\big(x, y^*_{\lambda,t}(x)\big)\right\| \;\leq\; \frac{C_f}{\lambda\,\mu}.$$

□

Lemma C.4. *Under Assumption 1, for any $\lambda \geq 2L_f/\mu$, it holds that*

$$\left|\, L^*_{\lambda,t}(x) - \varphi_t(x)\,\right| \;\leq\; \frac{D_0}{\lambda}, \qquad \textit{where} \qquad D_0 := \Big(C_f + \frac{C_f L_g}{2\mu}\Big)\frac{C_f}{\mu} = \mathcal{O}(\ell\,\kappa^2).$$

Proof. By definition,

$$\left|L^*_{\lambda,t}(x) - \varphi_t(x)\right| = \left|\, f_t\big(x, y^*_{\lambda,t}(x)\big) - f_t\big(x, y^*_t(x)\big)\,\right| \;+\; \lambda \left|\, g_t\big(x, y^*_{\lambda,t}(x)\big) - g_t\big(x, y^*_t(x)\big)\,\right|.$$

Since f_t is C_f-Lipschitz in y,

$$\left|\, f_t\big(x, y^*_{\lambda,t}(x)\big) - f_t\big(x, y^*_t(x)\big)\,\right| \;\leq\; C_f\left\|y^*_{\lambda,t}(x) - y^*_t(x)\right\|.$$

Because $\nabla_y g_t$ is L_g-Lipschitz and $\nabla_y g_t\big(x, y^*_t(x)\big) = 0$, the descent lemma gives

$$\left|\, g_t\big(x, y^*_{\lambda,t}(x)\big) - g_t\big(x, y^*_t(x)\big)\,\right| \;\leq\; \frac{L_g}{2}\left\|y^*_{\lambda,t}(x) - y^*_t(x)\right\|^2.$$

Combining the two bounds yields

$$\left|L^*_{\lambda,t}(x) - \varphi_t(x)\right| \;\le\; C_f \left\|y^*_{\lambda,t}(x) - y^*_t(x)\right\| \;+\; \lambda \frac{L_g}{2} \left\|y^*_{\lambda,t}(x) - y^*_t(x)\right\|^2.$$

By Lemma C.3, $\|y^*_{\lambda,t}(x) - y^*_t(x)\| \le C_f/(\lambda\mu)$. Substituting this,

$$\left|L^*_{\lambda,t}(x) - \varphi_t(x)\right| \;\le\; C_f \cdot \frac{C_f}{\lambda\mu} \;+\; \lambda \cdot \frac{L_g}{2} \cdot \left(\frac{C_f}{\lambda\mu}\right)^2 \;=\; \frac{1}{\lambda}\left(C_f + \frac{C_f L_g}{2\mu}\right)\frac{C_f}{\mu}.$$

This is exactly the claimed bound with $D_0 := \left(C_f + \frac{C_f L_g}{2\mu}\right)\frac{C_f}{\mu} = \mathcal{O}(\ell\,\kappa^2)$, where $\ell = \max\{C_f, L_f, L_g, \rho_g\}$ and $\kappa = \ell/\mu$. □

Lemma C.5. *Under Assumption 1, for $\lambda \ge 2L_f/\mu$, it holds that*

$$\left\|\nabla L^*_{\lambda,t}(x) - \nabla\varphi_t(x)\right\| \;\le\; \frac{D_1}{\lambda},$$

where

$$D_1 \;:=\; \left(L_f + \frac{\rho_g L_g}{\mu} + \frac{C_f L_g \rho_g}{2\mu^2} + \frac{C_f \rho_g}{2\mu}\right)\frac{C_f}{\mu} \;=\; \mathcal{O}(\ell\,\kappa^3).$$

Proof. Taking total derivative of $\varphi_t(x) = f_t\big(x, y^*_t(x)\big)$ gives

$$\nabla\varphi_t(x) = \nabla_x f_t\big(x, y^*_t(x)\big) - \nabla^2_{xy} g_t\big(x, y^*_t(x)\big)\,[\nabla^2_{yy} g_t\big(x, y^*_t(x)\big)]^{-1}\,\nabla_y f_t\big(x, y^*_t(x)\big).$$

Also,

$$\nabla L^*_{\lambda,t}(x) = \nabla_x f_t\big(x, y^*_{\lambda,t}(x)\big) + \lambda\Big(\nabla_x g_t\big(x, y^*_{\lambda,t}(x)\big) - \nabla_x g_t\big(x, y^*_t(x)\big)\Big).$$

Hence,

$$\begin{aligned}
\nabla L^*_{\lambda,t}(x) - \nabla\varphi_t(x) = {} & \underbrace{\nabla_x f_t\big(x, y^*_{\lambda,t}(x)\big) - \nabla_x f_t\big(x, y^*_t(x)\big)}_{\text{(A)}} \\
& + \underbrace{\nabla^2_{xy} g_t\big(x, y^*_t(x)\big)\,[\nabla^2_{yy} g_t\big(x, y^*_t(x)\big)]^{-1}\Big(\nabla_y f_t\big(x, y^*_t(x)\big) - \nabla_y f_t\big(x, y^*_{\lambda,t}(x)\big)\Big)}_{\text{(B)}} \\
& + \underbrace{\lambda\,\nabla^2_{xy} g_t\big(x, y^*_t(x)\big)\,[\nabla^2_{yy} g_t\big(x, y^*_t(x)\big)]^{-1}\Big(\nabla^2_{yy} g_t\big(x, y^*_t(x)\big)\,(y^*_{\lambda,t}(x) - y^*_t(x))}_{\text{(C)}} \\
& \qquad \underbrace{+\nabla_y g_t\big(x, y^*_t(x)\big) - \nabla_y g_t\big(x, y^*_{\lambda,t}(x)\big)\Big)}_{\text{(C, cont.)}} \\
& + \underbrace{\lambda\Big(\nabla_x g_t\big(x, y^*_{\lambda,t}(x)\big) - \nabla_x g_t\big(x, y^*_t(x)\big) - \nabla^2_{xy} g_t\big(x, y^*_t(x)\big)\,(y^*_{\lambda,t}(x) - y^*_t(x))\Big)}_{\text{(D)}}.
\end{aligned}$$

Taking norms and using the smoothness/curvature constants in Assumption 1,

$$\|\text{(A)}\| \le L_f\|y^*_{\lambda,t}(x) - y^*_t(x)\|, \quad \|\text{(B)}\| \le \frac{\rho_g L_g}{\mu}\|y^*_{\lambda,t}(x) - y^*_t(x)\|.$$

For the quadratic remainders,

$$\|\text{(C)}\| \le \frac{\lambda L_g \rho_g}{2\mu}\|y^*_{\lambda,t}(x) - y^*_t(x)\|^2, \qquad \|\text{(D)}\| \le \frac{\lambda\rho_g}{2}\|y^*_{\lambda,t}(x) - y^*_t(x)\|^2.$$

By Lemma C.3, $\|y^*_{\lambda,t}(x) - y^*_t(x)\| \le C_f/(\lambda\mu)$. Substituting this bound into the four terms above yields

$$\|\nabla L^*_{\lambda,t}(x) - \nabla\varphi_t(x)\| \;\le\; \left(L_f + \frac{\rho_g L_g}{\mu} + \frac{C_f L_g \rho_g}{2\mu^2} + \frac{C_f \rho_g}{2\mu}\right)\frac{C_f}{\lambda\mu} = \frac{D_1}{\lambda},$$

which proves the claim. □

Lemma C.6. *Under Assumption 1, for $\lambda \geq 2L_f/\mu$, it holds that*

$$\left\|\nabla y_t^*(x) - \nabla y_{\lambda,t}^*(x)\right\| \;\leq\; \frac{D_2}{\lambda}, \qquad D_2 \;:=\; \Big(\frac{1}{\mu} + \frac{2L_g}{\mu^2}\Big)\Big(L_f + \frac{C_f\rho_g}{\mu}\Big) \;=\; \mathcal{O}(\kappa^3).$$

Proof. Differentiating the first-order condition $\nabla_y g_t\big(x, y_t^*(x)\big) = 0$ w.r.t. x gives

$$\nabla_{xy}^2 g_t\big(x, y_t^*(x)\big) \;+\; \nabla y_t^*(x)\, \nabla_{yy}^2 g_t\big(x, y_t^*(x)\big) \;=\; 0. \tag{10}$$

Rearranging yields the closed form

$$\nabla y_t^*(x) \;=\; -\,\nabla_{xy}^2 g_t\big(x, y_t^*(x)\big)\big[\nabla_{yy}^2 g_t\big(x, y_t^*(x)\big)\big]^{-1}. \tag{11}$$

Similarly, since $y_{\lambda,t}^*(x)$ minimizes $L_{\lambda,t}(x,\cdot) = f_t(x,\cdot) + \lambda g_t(x,\cdot)$,

$$\nabla y_{\lambda,t}^*(x) \;=\; -\,\nabla_{xy}^2 L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big)\big[\nabla_{yy}^2 L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big)\big]^{-1}. \tag{12}$$

Using the identity $A^{-1} - B^{-1} = A^{-1}(B-A)B^{-1}$ and the strong convexity in y, we first bound the difference of the inverses:

$$\left\|\big[\nabla_{yy}^2 g_t(x, y_t^*(x))\big]^{-1} - \Big[\frac{1}{\lambda}\nabla_{yy}^2 L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big)\Big]^{-1}\right\| \;\leq\; \frac{2}{\lambda\mu^2}\Big(L_f + \frac{C_f\rho_g}{\mu}\Big),$$

where we used $\|\nabla_{yy}^2 f_t\| \leq L_f$, the ρ_g-Lipschitzness of $\nabla^2 g_t$ in y, and Lemma C.3: $\|y_{\lambda,t}^*(x) - y_t^*(x)\| \leq C_f/(\lambda\mu)$.

Now normalize Equations (10) and (11) by λ for comparison and apply the triangle inequality:

$$\begin{aligned}\|\nabla y_{\lambda,t}^*(x) - \nabla y_t^*(x)\| \leq{}& \left\|\nabla_{xy}^2 g_t\big(x, y_t^*(x)\big) - \frac{1}{\lambda}\nabla_{xy}^2 L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big)\right\| \cdot \left\|[\nabla_{yy}^2 g_t(x, y_t^*(x))]^{-1}\right\| \\ &+ \left\|\frac{1}{\lambda}\nabla_{xy}^2 L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big)\right\| \cdot \left\|[\nabla_{yy}^2 g_t(x, y_t^*(x))]^{-1} - \Big[\frac{1}{\lambda}\nabla_{yy}^2 L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big)\Big]^{-1}\right\|.\end{aligned}$$

The first factor is bounded by

$$\frac{1}{\mu}\Big(\frac{L_f}{\lambda} + \rho_g\|y_{\lambda,t}^*(x) - y_t^*(x)\|\Big) \;\leq\; \frac{1}{\lambda\mu}\Big(L_f + \frac{C_f\rho_g}{\mu}\Big),$$

using again Lemma C.3. For the second factor, from $\lambda \geq 2L_f/\mu$ we have $\big\|\frac{1}{\lambda}\nabla_{xy}^2 L_{\lambda,t}(\cdot,\cdot)\big\| \leq 2L_g$, which combined with equation 12 yields

$$\left\|\frac{1}{\lambda}\nabla_{xy}^2 L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big)\right\| \cdot \left\|[\nabla_{yy}^2 g_t(\cdot)]^{-1} - \Big[\tfrac{1}{\lambda}\nabla_{yy}^2 L_{\lambda,t}(\cdot)\Big]^{-1}\right\| \;\leq\; \frac{2L_g}{\lambda\mu^2}\Big(L_f + \frac{C_f\rho_g}{\mu}\Big).$$

Putting the two bounds together gives

$$\|\nabla y_{\lambda,t}^*(x) - \nabla y_t^*(x)\| \;\leq\; \Big(\frac{1}{\lambda\mu} + \frac{2L_g}{\lambda\mu^2}\Big)\Big(L_f + \frac{C_f\rho_g}{\mu}\Big) \;=\; \frac{D_2}{\lambda},$$

which proves the claim. $\square$

Lemma C.7. *Under Assumption 1, for $\lambda \geq 2L_f/\mu$, it holds that*

$$\left\|\nabla y_{\lambda,t}^*(x)\right\| \;\leq\; \frac{4L_g}{\mu}.$$

Proof. Since $y_{\lambda,t}^*(x)$ minimizes $L_{\lambda,t}(x,\cdot) = f_t(x,\cdot) + \lambda g_t(x,\cdot)$, the first-order optimality condition $\nabla_y L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big) = 0$ and the implicit function theorem give

$$\nabla y_{\lambda,t}^*(x) = -\,\nabla_{xy}^2 L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big)\,\Big[\nabla_{yy}^2 L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big)\Big]^{-1}.$$

Taking operator norms yields

$$\left\|\nabla y_{\lambda,t}^*(x)\right\| \;\leq\; \left\|\nabla_{xy}^2 L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big)\right\| \, \left\|\Big[\nabla_{yy}^2 L_{\lambda,t}\big(x, y_{\lambda,t}^*(x)\big)\Big]^{-1}\right\|.$$

By smoothness, $\|\nabla^2_{xy} L_{\lambda,t}(\cdot,\cdot)\| \le \|\nabla^2_{xy} f_t(\cdot,\cdot)\| + \lambda\|\nabla^2_{xy} g_t(\cdot,\cdot)\| \le L_f + \lambda L_g \le 2\lambda L_g$ when $\lambda \ge 2L_f/\mu$. Moreover, by Lemma C.2, $L_{\lambda,t}(x,\cdot)$ is $(\lambda\mu/2)$-strongly convex, so $\nabla^2_{yy} L_{\lambda,t}(\cdot,\cdot) \succeq (\lambda\mu/2)I$ and hence $\big\|\big[\nabla^2_{yy} L_{\lambda,t}(\cdot,\cdot)\big]^{-1}\big\| \le 2/(\lambda\mu)$. Combining the two bounds gives

$$\big\|\nabla y^*_{\lambda,t}(x)\big\| \;\le\; (2\lambda L_g)\cdot \frac{2}{\lambda\mu} \;=\; \frac{4L_g}{\mu}.$$

□

Lemma C.8 (Smoothness of $L^*_{\lambda,t}(x)$)**.** *Under Assumption 1, for $\lambda \ge 2L_f/\mu$, the mapping $\nabla L^*_{\lambda,t}(x)$ is D_3-Lipschitz, where*

$$D_3 \;:=\; L_f \;+\; \frac{4L_f L_g}{\mu} \;+\; \frac{C_f\rho_g}{\mu} \;+\; \frac{C_f L_g \rho_g}{\mu^2} \;+\; L_g D_2 \;=\; \mathcal{O}(\ell\kappa^3),$$

and D_2 is the constant from Lemma C.6.

Proof. Decompose

$$\nabla L^*_{\lambda,t}(x) = \underbrace{\nabla_x f_t\big(x, y^*_{\lambda,t}(x)\big)}_{=:A(x)} \;+\; \underbrace{\lambda\Big(\nabla_x g_t\big(x, y^*_{\lambda,t}(x)\big) - \nabla_x g_t\big(x, y^*_t(x)\big)\Big)}_{=:B(x)}. \tag{13}$$

By Lemma C.7, $y^*_{\lambda,t}(x)$ is $(4L_g/\mu)$-Lipschitz. Therefore $A(x)$ is $\big(1+\frac{4L_g}{\mu}\big)L_f$-Lipschitz (chain rule with the L_f-smoothness of f_t).

For $B(x)$, take the total derivative:

$$\begin{aligned}\nabla B(x) = {} & \lambda\big(\nabla^2_{xx} g_t(x, y^*_{\lambda,t}(x)) - \nabla^2_{xx} g_t(x, y^*_t(x))\big) \quad (14)\\ & + \lambda\big(\nabla y^*_{\lambda,t}(x)\,\nabla^2_{yx} g_t(x, y^*_{\lambda,t}(x)) - \nabla y^*_t(x)\,\nabla^2_{yx} g_t(x, y^*_t(x))\big). \quad (15)\end{aligned}$$

Hence

$$\begin{aligned}\|\nabla B(x)\| & \le \lambda\|\nabla^2_{xx} g_t(x, y^*_{\lambda,t}) - \nabla^2_{xx} g(x, y^*_t)\| + \lambda\|\nabla y^*_t(x)\|\,\|\nabla^2_{yx} g_t(x, y^*_{\lambda,t}) - \nabla^2_{yx} g_t(x, y^*_t)\| \quad (16)\\ & \quad + \lambda\|\nabla y^*_{\lambda,t}(x) - \nabla y^*_t(x)\|\,\|\nabla^2_{yx} g_t(x, y^*_{\lambda,t})\| \quad (17)\\ & \le \lambda\rho_g\left(1+\frac{L_g}{\mu}\right)\|y^*_{\lambda,t}(x) - y^*_t(x)\| + \lambda L_g\,\|\nabla y^*_{\lambda,t}(x) - \nabla y^*_t(x)\|, \quad (18)\end{aligned}$$

where we used $\|\nabla y^*_t(x)\| \le L_g/\mu$ and the y-Lipschitzness (ρ_g) of $\nabla^2 g_t$, together with $\|\nabla^2_{yx} g_t\| \le L_g$.

Applying Lemma C.3, $\|y^*_{\lambda,t}(x) - y^*_t(x)\| \le C_f/(\lambda\mu)$, and Lemma C.6, $\|\nabla y^*_{\lambda,t}(x) - \nabla y^*_t(x)\| \le D_2/\lambda$, we obtain

$$\|\nabla B(x)\| \;\le\; \left(1+\frac{L_g}{\mu}\right)\frac{C_f\rho_g}{\mu} \;+\; L_g D_2,$$

so $B(x)$ is $\big((1+L_g/\mu)\,C_f\rho_g/\mu + L_g D_2\big)$-Lipschitz.

Combining the Lipschitz constants of A and B yields

$$D_3 \;=\; \left(1+\frac{4L_g}{\mu}\right)L_f \;+\; \left(1+\frac{L_g}{\mu}\right)\frac{C_f\rho_g}{\mu} \;+\; L_g D_2 \;=\; L_f + \frac{4L_f L_g}{\mu} + \frac{C_f\rho_g}{\mu} + \frac{C_f L_g\rho_g}{\mu^2} + L_g D_2,$$

which proves the claim. □

Lemma C.9. *Let $L_\phi = \mathcal{O}(\kappa^3\ell)$ be a uniform gradient Lipschitz constant of ϕ_t [Chen et al., 2024, Lemma 4.4]. Assume $\eta^x_t \le \frac{1}{2L_\phi}$, and define the (fully first-order) proxy direction*

$$q^x_t := \nabla_x f_t(x_t, y_t) + \lambda_t\big(\nabla_x g_t(x_t, y_t) - \nabla_x g_t(x_t, z_t)\big).$$

Let $e^y_t := \|y_t - y^*_{\lambda_t,t}(x_t)\|$ and $e^z_t := \|z_t - y^*_t(x_t)\|$. Then

$$\begin{aligned}\phi_t(x_{t+1}) \le {} & \phi_t(x_t) - \frac{1}{4\eta^x_t}\|x_{t+1} - x_t\|^2 - \frac{\eta^x_t}{2}\|\nabla\phi_t(x_t)\|^2\\ & + \frac{\eta^x_t D_1^2}{\lambda_t^2} + 2\eta^x_t(L_f + \lambda_t L_g)^2\,(e^y_t)^2 + 2\eta^x_t(\lambda_t L_g)^2\,(e^z_t)^2. \quad (19)\end{aligned}$$

Proof. From L_ϕ-smoothness of ϕ_t, we have

$$\begin{aligned}
\phi_t(x_{t+1}) - \phi_t(x_t) &\le -\eta_t^x \langle \nabla \phi_t(x_t), q_t^x \rangle + \frac{L_\phi}{2}\|x_{t+1} - x_t\|^2 \\
&\le -\frac{\eta_t^x}{2}\left(\|\nabla \phi_t(x_t)\|^2 + \|q_t^x\|^2 - \|\nabla \phi_t(x_t) - q_t^x\|^2\right) + \frac{L_\phi}{2}\|x_{t+1} - x_t\|^2 \\
&\le -\frac{\eta_t^x}{2}\|\nabla \phi_t(x_t)\|^2 - \left(\frac{1}{2\eta_t^x} - \frac{L_\phi}{2}\right)\|x_{t+1} - x_t\|^2 + \frac{\eta_t^x}{2}\|\nabla \phi_t(x_t) - q_t^x\|^2 \\
&\le -\frac{\eta_t^x}{2}\|\nabla \phi_t(x_t)\|^2 - \frac{1}{4\eta_t^x}\|x_{t+1} - x_t\|^2 + \frac{\eta_t^x}{2}\|\nabla \phi_t(x_t) - q_t^x\|^2, \qquad (20)
\end{aligned}$$

where the last inequality comes from $\eta_t^x \le \frac{1}{2L_\phi}$. Moreover,

$$\begin{aligned}
\|\nabla \phi_t(x_t) - q_t^x\|^2 &\le 2\|\nabla \phi_t(x_t) - \nabla L^*_{\lambda_t,t}(x_t)\|^2 + 2\|\nabla L^*_{\lambda_t,t}(x_t) - q_t^x\|^2 \\
&\le \frac{2D_1^2}{\lambda_t^2} + 2\|\nabla L^*_{\lambda_t,t}(x_t) - q_t^x\|^2, \qquad (21)
\end{aligned}$$

where the last inequality follows by Lemma C.5 and $D_1 = \mathcal{O}(\kappa^3\ell)$. The estimator error $\|\nabla L^*_{\lambda_t,t}(x_t) - q_t^x\|^2$ is controlled by the inner-loop errors:

$$\|\nabla L^*_{\lambda_t,t}(x_t) - q_t^x\|^2 \le 2(L_f + \lambda_t L_g)^2 \cdot \|y_t - y^*_{\lambda_t,t}(x_t)\|^2 + 2\lambda_t^2 L_g^2 \|z_t - y_t^*(x_t)\|^2. \qquad (22)$$

Combine the last three displays and simplify constants to obtain the claim. □

Lemma C.10 (Inexact descent on an L-smooth proxy)**.** *Let L be the gradient Lipschitz constant of $L^*_{\lambda,t}(x)$ derived in Lemma C.8. Let $e_t^y := \left\|y_t - y^*_{\lambda,t}(x_t)\right\|$ and $e_t^z := \|z_t - y_t^*(x_t)\|$. If $\eta_t^x \le 1/L$, then under Assumption 1, we have*

$$L^*_{\lambda,t}(x_{t+1}) \;\le\; L^*_{\lambda,t}(x_t) \;-\; \frac{\eta_t^x}{2}\,\left\|\nabla L^*_{\lambda,t}(x_t)\right\|^2 \;+\; \frac{\eta_t^x}{2}\Big((L_f + \lambda L_g)e_t^y + \lambda L_g e_t^z\Big)^2.$$

Proof. By Lemma C.8 and our choice of λ, we have $L := \sup_x \|\nabla^2 L^*_\lambda(x)\| = O(\ell\kappa^3)$. Consider one outer step of Algorithm 1: $x_{t+1} = x_t - \eta_t^x\,\widehat{\nabla L}_t$, where

$$\widehat{\nabla L}_t = \nabla_x f_t(x_t, y_t) + \lambda\big(\nabla_x g_t(x_t, y_t) - \nabla_x g_t(x_t, z_t)\big).$$

By L-smoothness of $L^*_{\lambda,t}$ and letting $\eta_t^x \le \frac{1}{L}$,

$$\begin{aligned}
L^*_{\lambda,t}(x_{t+1}) &\le L^*_{\lambda,t}(x_t) + \big\langle \nabla L^*_{\lambda,t}(x_t),\, x_{t+1} - x_t \big\rangle + \frac{L}{2}\|x_{t+1} - x_t\|^2 \qquad (23)\\
&= L^*_{\lambda,t}(x_t) - \frac{\eta_t^x}{2}\|\nabla L^*_{\lambda,t}(x_t)\|^2 - \Big(\frac{\eta_t^x}{2} - \frac{(\eta_t^x)^2 L}{2}\Big)\left\|\widehat{\nabla L}_t\right\|^2 + \frac{\eta_t^x}{2}\|\widehat{\nabla L}_t - \nabla L^*_{\lambda,t}(x_t)\|^2 \qquad (24)\\
& \qquad (25)
\end{aligned}$$

The estimator error is controlled by the inner-loop errors:

$$\|\widehat{\nabla L}_t - \nabla L^*_{\lambda,t}(x_t)\| \le (L_f + \lambda L_g) \cdot \|y_t - y^*_{\lambda,t}(x_t)\| + \lambda L_g \|z_t - y_t^*(x_t)\|. \qquad (26)$$

Since $\eta_t^x \le 1/L$, we have $\frac{\eta_t^x}{2} - \frac{(\eta_t^x)^2 L}{2} \ge 0$, hence we can drop the corresponding nonpositive term in equation 23. Combining equation 23 and equation 26 and using the definitions of e_t^y, e_t^z yields the claim. □

D PROOFS FOR DETERMINISTIC ANALYSIS

We next derive a one-step recursion for the inner tracking errors $\|y_t - y^*_{\lambda,t}(x_t)\|$ and $\|z_t - y^*_t(x_t)\|$ in terms of (i) the outer motion $\|x_{t+1} - x_t\|$ and (ii) drift of the follower solution map.

Lemma D.1 (One-step contraction of inner errors with drift)**.** *Fix $\lambda_t \geq 2L_f/\mu$ and step-sizes $\eta^y_t \leq \frac{1}{L_f+\lambda_t L_g}$ and $\eta^z_t \leq \frac{1}{L_g}$. Let*

$$e^y_t := \left\|y_t - y^*_{\lambda,t}(x_t)\right\|, \quad e^z_t := \left\|z_t - y^*_t(x_t)\right\|, \quad \delta_t := (e^y_t)^2 + (e^z_t)^2.$$

Define the (unpenalized) target drift

$$\Gamma_t := \sup_x \left\|y^*_{t+1}(x) - y^*_t(x)\right\|^2,$$

and the constants

$$c^t_y := \frac{\lambda_t \mu}{2}\eta^y_t, \qquad c^t_z := \mu \eta^z_t, \qquad c^t_0 := \tfrac{1}{2}\min\{c^t_y, c^t_z\}.$$

Then under Assumption 1, there exist absolute constants

$$K^t_y := 1 + \frac{2(1-c^t_y)}{c^t_y} \leq \frac{3}{c^t_y}, \qquad K^t_z := 1 + \frac{2(1-c^t_z)}{c^t_z} \leq \frac{3}{c^t_z},$$

such that, with $\Delta x_t := x_{t+1} - x_t$,

$$\delta_{t+1} \;\leq\; (1-c^t_0)\,\delta_t + \underbrace{\left(\tfrac{32L_g^2}{\mu^2}K^t_y + \tfrac{2L_g^2}{\mu^2}K^t_z\right)}_{=:C^t_x}\|x_{t+1}-x_t\|^2 + (6K^t_y + 2K^t_z)\,\Gamma_t \;+ K^t_y\left[\frac{6C_f^2}{\lambda_t^2\mu^2} + \frac{6C_f^2}{\lambda_{t+1}^2\mu^2}\right]. \tag{27}$$

Proof. We will use the linear contraction lemma in the following:

Lemma D.2 (Bubeck et al. [2015], Theorem 3.1)**.** *Suppose $h : \mathbb{R}^d \to \mathbb{R}$ is β-gradient Lipschitz and α-strongly convex. Consider gradient descent with step size $\eta \leq 1/\beta$:*

$$x_{t+1} \;=\; x_t \;-\; \eta\,\nabla h(x_t).$$

Let $x^* = \arg\min_{x\in\mathbb{R}^d} h(x)$. Then

$$\|x_{t+1} - x^*\|^2 \;\leq\; \Big(1-\eta\alpha\Big)\,\|x_t - x^*\|^2.$$

Step 1: Contraction towards the *current* targets. By Lemma C.2, $y \mapsto L_{\lambda_t,t}(x_t, y)$ is $(\lambda_t\mu/2)$-strongly convex, and it is $(L_f + \lambda_t L_g)$-smooth; gradient descent with $\eta^y_t \leq 1/(L_f + \lambda_t L_g)$ yields

$$\|y_{t+1} - y^*_{\lambda,t}(x_t)\|^2 \;\leq\; (1-c^t_y)\,\|y_t - y^*_{\lambda,t}(x_t)\|^2, \quad c^t_y = \frac{\lambda_t\mu}{2}\eta^y_t.$$

Similarly, since $y \mapsto g_t(x_t, y)$ is μ-strongly convex and L_g-smooth, the update $z_{t+1} = z_t - \eta^z_t \nabla_y g_t(x_t, z_t)$ with $\eta^z_t \leq 1/L_g$ gives

$$\|z_{t+1} - y^*_t(x_t)\|^2 \;\leq\; (1-c^t_z)\,\|z_t - y^*_t(x_t)\|^2, \quad c^t_z = \mu\eta^z_t.$$

Step 2: Weighted Young's inequality. For any $\theta > 0$, $\|a+b\|^2 \leq (1+\theta)\|a\|^2 + (1+1/\theta)\|b\|^2$. Apply this with

$$a = y_{t+1} - y^*_{\lambda,t}(x_t), \quad b = y^*_{\lambda,t}(x_t) - y^*_{\lambda,t+1}(x_{t+1}),$$

and choose $\theta^t_y := \frac{c^t_y}{2(1-c^t_y)}$ so that $(1+\theta^t_y)(1-c^t_y) = 1 - \frac{c^t_y}{2}$:

$$\begin{aligned}
&\|y_{t+1} - y^*_{\lambda,t+1}(x_{t+1})\|^2 \\
&\qquad \leq (1+\theta^t_y)\,\|y_{t+1} - y^*_{\lambda,t}(x_t)\|^2 + \left(1 + \frac{1}{\theta^t_y}\right)\cdot\|y^*_{\lambda,t+1}(x_{t+1}) - y^*_{\lambda,t}(x_t)\|^2 \\
&\qquad \leq\; (1-\tfrac{c^t_y}{2})\,\|y_t - y^*_{\lambda,t}(x_t)\|^2 \;+\; \underbrace{\left(1+\tfrac{1}{\theta^t_y}\right)}_{=\,K^t_y}\,\|y^*_{\lambda,t+1}(x_{t+1}) - y^*_{\lambda,t}(x_t)\|^2.
\end{aligned} \tag{28}$$

The same argument for z with $\theta_z^t := \frac{c_z^t}{2(1-c_z^t)}$ gives

$$\|z_{t+1} - y_{t+1}^*(x_{t+1})\|^2 \;\le\; (1 - \tfrac{c_z^t}{2})\, \|z_t - y_t^*(x_t)\|^2 \;+\; K_z^t\, \|y_{t+1}^*(x_{t+1}) - y_t^*(x_t)\|^2, \quad K_z^t = 1 + \tfrac{1}{\theta_z^t}. \tag{29}$$

Step 3: Lipschitzness of $y_t^*(\cdot)$ and $y_{\lambda,t}^*(\cdot)$. By Lemma C.1, $y_t^*(\cdot)$ and $y_{\lambda,t}^*(\cdot)$ are Lipschitz in x with constants L_g/μ and $4L_g/\mu$, respectively. From L_g/μ-Lipschitzness of $y_t^*(x)$, we have

$$\begin{aligned}\|y_{t+1}^*(x_{t+1}) - y_t^*(x_t)\|^2 &\le 2\|y_{t+1}^*(x_{t+1}) - y_{t+1}^*(x_t)\|^2 + 2\|y_{t+1}^*(x_t) - y_t^*(x_t)\|^2 \\ &\le \frac{2L_g^2}{\mu^2} \cdot \|x_{t+1} - x_t\|^2 + 2 \sup_x \|y_{t+1}^*(x) - y_t^*(x)\|^2, \end{aligned} \tag{30}$$

and similarly from $4L_g/\mu$-Lipschitzness of $y_{\lambda,t}^*(x)$, we get

$$\begin{aligned}\|y_{\lambda_{t+1},t+1}^*(x_{t+1}) - y_{\lambda_t,t}^*(x_t)\|^2 &\le 2\|y_{\lambda_{t+1},t+1}^*(x_{t+1}) - y_{\lambda_{t+1},t+1}^*(x_t)\|^2 + 2\|y_{\lambda_{t+1},t+1}^*(x_t) - y_{\lambda_t,t}^*(x_t)\|^2 \\ &\le \frac{32L_g^2}{\mu^2} \cdot \|x_{t+1} - x_t\|^2 + 2\|y_{\lambda_{t+1},t+1}^*(x_t) - y_{\lambda_t,t}^*(x_t)\|^2, \\ &\overset{(a)}{\le} \frac{32L_g^2}{\mu^2} \cdot \|x_{t+1} - x_t\|^2 + 6\|y_{\lambda_{t+1},t+1}^*(x_t) - y_{t+1}^*(x_t)\|^2 + 6\|y_{t+1}^*(x_t) - y_t^*(x_t)\|^2 + 6\|y_{\lambda_t,t}^*(x_t) - y_t^*(x_t)\|^2 \\ &\overset{(b)}{\le} \frac{32L_g^2}{\mu^2} \cdot \|x_{t+1} - x_t\|^2 + 6\|y_{t+1}^*(x_t) - y_t^*(x_t)\|^2 + \frac{6C_f^2}{\lambda_t^2\mu^2} + \frac{6C_f^2}{\lambda_{t+1}^2\mu^2} \\ &\le \frac{32L_g^2}{\mu^2} \cdot \|x_{t+1} - x_t\|^2 + 6 \sup_x \|y_{t+1}^*(x) - y_t^*(x)\|^2 + \frac{6C_f^2}{\lambda_t^2\mu^2} + \frac{6C_f^2}{\lambda_{t+1}^2\mu^2}, \end{aligned} \tag{31}$$

where (a) follows from $(a+b+c)^2 \le 3(a^2+b^2+c^2)$ and (b) comes from Lemma C.3.

Plugging equation 31 into equation 28, we get

$$\begin{aligned}\|y_{t+1} - y_{\lambda,t+1}^*(x_{t+1})\|^2 \;&\le\; (1 - \tfrac{c_y^t}{2})\, \|y_t - y_{\lambda,t}^*(x_t)\|^2 \\ &+ \underbrace{\left(1 + \tfrac{1}{\theta_y^t}\right)}_{= K_y^t} \left[\frac{32L_g^2}{\mu^2} \cdot \|x_{t+1} - x_t\|^2 + 6 \sup_x \|y_{t+1}^*(x) - y_t^*(x)\|^2 + \frac{6C_f^2}{\lambda_t^2\mu^2} + \frac{6C_f^2}{\lambda_{t+1}^2\mu^2}\right]. \end{aligned} \tag{32}$$

And similarly, by plugging equation 30 into equation 29 with $K_z^t = 1 + \frac{1}{\theta_z^t}$, we have

$$\|z_{t+1} - y_{t+1}^*(x_{t+1})\|^2 \;\le\; (1 - \tfrac{c_z^t}{2})\, \|z_t - y_t^*(x_t)\|^2 \;+\; K_z^t \left[\frac{2L_g^2}{\mu^2} \cdot \|x_{t+1} - x_t\|^2 + 2 \sup_x \|y_{t+1}^*(x) - y_t^*(x)\|^2\right]. \tag{33}$$

Finally, by adding equation 32 and equation 33 and setting $c_0^t := \frac{1}{2}\min\{c_y^t, c_z^t\}$ yields

$$\delta_{t+1} \;\le\; (1 - c_0^t)\, \delta_t + \left(\tfrac{32L_g^2}{\mu^2} K_y^t + \tfrac{2L_g^2}{\mu^2} K_z^t\right) \|x_{t+1} - x_t\|^2 + (6K_y^t + 2K_z^t)\, \Gamma_t \;+ K_y^t \left[\frac{6C_f^2}{\lambda_t^2\mu^2} + \frac{6C_f^2}{\lambda_{t+1}^2\mu^2}\right],$$

as claimed. □

D.1 PROOF OF THEOREM 4.1

Proof of Theorem 4.1. For clarity we present the proof for the (common) choice of *constant* parameters: fix $\lambda \ge 2L_f/\mu$ and stepsizes

$$\eta_t^y = \eta^y \le \frac{1}{L_f + \lambda L_g}, \qquad \eta_t^z = \eta^z \le \frac{1}{L_g}, \qquad \eta_t^x = \eta^x \le \frac{1}{2L_\phi} \quad (t = 1, \ldots, T),$$

and let (x_t, y_t, z_t) be generated by Algorithm 1 with the round index convention "use (x_t, y_t, z_t) at round t and update to $(x_{t+1}, y_{t+1}, z_{t+1})$". Equivalently, if the algorithm is initialized at (x_0, y_0, z_0), one may set $(x_1, y_1, z_1) := (x_0, y_0, z_0)$ so that the initial tracking error δ_1 matches the initialization term in Theorem 4.1. Write

$$e_t^y := \|y_t - y_{\lambda,t}^*(x_t)\|, \qquad e_t^z := \|z_t - y_t^*(x_t)\|, \qquad \delta_t := (e_t^y)^2 + (e_t^z)^2.$$

Step 1: Sum the outer descent inequality. Lemma C.9 gives, for each t,

$$\begin{aligned}\frac{\eta^x}{2}\|\nabla\phi_t(x_t)\|^2+\frac{1}{4\eta^x}\|x_{t+1}-x_t\|^2 &\le \phi_t(x_t)-\phi_t(x_{t+1})\frac{\eta^x D_1^2}{\lambda^2}2\eta^x(L_f+\lambda L_g)^2(e_t^y)^2 2\eta^x(\lambda L_g)^2(e_t^z)^2\\ &\le \phi_t(x_t)-\phi_t(x_{t+1})\frac{\eta^x D_1^2}{\lambda^2}\eta^x C_e\,\delta_t,\end{aligned} \tag{34}$$

where $C_e := 2\big((L_f+\lambda L_g)^2+(\lambda L_g)^2\big)$. Summing equation 34 over $t=1,\ldots,T$ yields

$$\frac{\eta^x}{2}\sum_{t=1}^T\|\nabla\phi_t(x_t)\|^2\frac{1}{4\eta^x}\sum_{t=1}^T\|x_{t+1}-x_t\|^2\le\sum_{t=1}^T\big(\phi_t(x_t)-\phi_t(x_{t+1})\big)\frac{\eta^x T D_1^2}{\lambda^2}\eta^x C_e\sum_{t=1}^T\delta_t. \tag{35}$$

Step 2: Telescope the nonstationary outer terms via budgets. By reindexing,

$$\sum_{t=1}^T\big(\phi_t(x_t)-\phi_t(x_{t+1})\big) \;=\; \phi_1(x_1)-\phi_T(x_{T+1})+\sum_{t=2}^T\big(\phi_t(x_t)-\phi_{t-1}(x_t)\big).$$

Since $\phi_T(x_{T+1})\ge\inf_x\phi_T(x)$, it remains to bound the variation term. For any x,

$$\begin{aligned}\big|\phi_t(x)-\phi_{t-1}(x)\big| &= \big|f_t(x,y_t^*(x))-f_{t-1}(x,y_{t-1}^*(x))\big|\\ &\le \underbrace{\big|f_t(x,y_t^*(x))-f_{t-1}(x,y_t^*(x))\big|}_{\le\sup_{x,y}|f_t(x,y)-f_{t-1}(x,y)|}+\underbrace{\big|f_{t-1}(x,y_t^*(x))-f_{t-1}(x,y_{t-1}^*(x))\big|}_{\le C_f\|y_t^*(x)-y_{t-1}^*(x)\|}.\end{aligned}$$

Taking $\sup_x$ and summing over $t=2,\ldots,T$ gives

$$\sum_{t=2}^T\sup_x\big|\phi_t(x)-\phi_{t-1}(x)\big|\le V_T^f+C_f H_{1,T}. \tag{36}$$

Therefore,

$$\sum_{t=1}^T\big(\phi_t(x_t)-\phi_t(x_{t+1})\big)\le\phi_1(x_1)-\inf_x\phi_T(x)+V_T^f+C_fH_{1,T}. \tag{37}$$

Step 3: Bound $\sum_t\delta_t$ from the inner tracking recursion. Lemma D.1 gives, for each t, a recursion of the form

$$\delta_{t+1}\le(1-c_0)\delta_t+A_x\|x_{t+1}-x_t\|^2+A_\Gamma\,\Gamma_t+A_\lambda\cdot\frac{1}{\lambda^2}, \tag{38}$$

where $c_0:=\inf_t c_0^t>0$ and the constants A_x, A_Γ, A_λ are polynomial in $(\mu^{-1},L_f,L_g,\rho_g)$ (they are explicit combinations of K_y^t, K_z^t in Lemma D.1). Summing equation 38 over $t=1,\ldots,T$ and using $\delta_{T+1}\ge0$ gives

$$\sum_{t=1}^T\delta_t\le\frac{\delta_1}{c_0}\frac{A_x}{c_0}\sum_{t=1}^T\|x_{t+1}-x_t\|^2\frac{A_\Gamma}{c_0}\sum_{t=1}^T\Gamma_t\frac{A_\lambda}{c_0}\cdot\frac{T}{\lambda^2}. \tag{39}$$

By the definition $\Gamma_t := \sup_x\|y_{t+1}^*(x)-y_t^*(x)\|^2$ in Lemma D.1, we have $\sum_{t=1}^{T-1}\Gamma_t = H_{2,T}$ and hence $\sum_{t=1}^T\Gamma_t\le H_{2,T}+\Gamma_T$. (One can define $\Gamma_T:=0$ to avoid the boundary term.)

Step 4: Absorb the $\sum_t\|x_{t+1}-x_t\|^2$ coupling and conclude. Plug equation 37 and equation 39 into equation 35 to obtain

$$\begin{aligned}\frac{\eta^x}{2}\sum_{t=1}^T\|\nabla\phi_t(x_t)\|^2\Big(\frac{1}{4\eta^x}-\eta^xC_e\frac{A_x}{c_0}\Big)\sum_{t=1}^T\|x_{t+1}-x_t\|^2 &\le \phi_1(x_1)-\inf_x\phi_T(x)+V_T^f+C_fH_{1,T}\\ &\quad+\frac{\eta^xTD_1^2}{\lambda^2}\eta^xC_e\Big(\frac{\delta_1}{c_0}+\frac{A_\Gamma}{c_0}H_{2,T}+\frac{A_\lambda}{c_0}\cdot\frac{T}{\lambda^2}\Big).\end{aligned} \tag{40}$$

Choose η^x small enough so that $\eta^x \le \sqrt{\frac{c_0}{8C_eA_x}}$, which makes the coefficient of $\sum_t \|x_{t+1}-x_t\|^2$ nonnegative. Dropping this nonnegative term from the left-hand side and dividing by $\eta^x T$ yields

$$\frac{1}{T}\sum_{t=1}^{T}\|\nabla\phi_t(x_t)\|^2 \le \frac{2}{\eta^x T}\Big(\phi_1(x_1) - \inf_x \phi_T(x) + V_T^f + C_f H_{1,T}\Big) \;+\; \frac{2D_1^2}{\lambda^2} \\ +\; \frac{2C_e}{c_0 T}\delta_1 \;+\; \frac{2C_eA_\Gamma}{c_0 T}H_{2,T} \;+\; \frac{2C_eA_\lambda}{c_0}\cdot\frac{1}{\lambda^2}. \tag{41}$$

This is an explicit deterministic local-regret bound in terms of $(V_T^f, H_{1,T}, H_{2,T})$, the proxy bias $1/\lambda^2$, and the initialization error δ_1. Finally, taking $\eta^x = \Theta(T^{-1/2})$ (subject to $\eta^x \le 1/(2L_\phi)$ and the absorption condition above) gives the $\tilde{O}(T^{-1/2})$ scaling claimed in Theorem 4.1. $\square$

For completeness, we also provide a fully detailed (windowed) deterministic analysis in Theorem D.1 below.

Lemma C.9 (windowed inexact descent using only ϕ-smoothness). Let $\{u_i\}_{i=0}^{w-1}$ be nonnegative window weights with sum $W = \sum_{i=0}^{w-1} u_i$, and define the averaged outer objective

$$\bar{\phi}_t(x) := \frac{1}{W}\sum_{i=0}^{w-1} u_i\,\phi_{t-i}(x).$$

Assume each ϕ_s is L_ϕ-smooth with the same constant L_ϕ (as in Appendix C, $L_\phi = O(\kappa^3\ell)$). Consider the outer update

$$x_{t+1} \;=\; x_t - \eta_t^x\,\bar{q}_t^x, \qquad \bar{q}_t^x \;:=\; \bar{\nabla}_x f_t(x_t,y_t) + \lambda\big(\bar{\nabla}_x g_t(x_t,y_t) - \bar{\nabla}_x g_t(x_t,z_t)\big),$$

where $\bar{\nabla}$ denotes the windowed partial gradients: $\bar{\nabla}_u h_t(\cdot) := \dfrac{1}{W}\displaystyle\sum_{i=0}^{w-1} u_i\,\nabla_u h_{t-i}(\cdot)$ for $u \in \{x,y\}$ and $h \in \{f,g\}$. If $\eta_t^x \le 1/(2L_\phi)$, then

$$\bar{\phi}_t(x_{t+1}) \;\le\; \bar{\phi}_t(x_t) \;-\; \frac{\eta_t^x}{2}\big\|\nabla\bar{\phi}_t(x_t)\big\|^2 \;-\; \frac{1}{4\,\eta_t^x}\|x_{t+1}-x_t\|^2 \;+\; \frac{\eta_t^x}{2}\big\|\nabla\bar{\phi}_t(x_t) - \bar{q}_t^x\big\|^2. \tag{42}$$

No smoothness of $L_{\lambda,t}^\star$ is used.

Proof. Each ϕ_{t-i} is L_ϕ-smooth, so its gradient is L_ϕ-Lipschitz. By linearity of gradients,

$$\nabla\bar{\phi}_t(x) = \frac{1}{W}\sum_{i=0}^{w-1} u_i\,\nabla\phi_{t-i}(x),$$

and for any x, y,

$$\|\nabla\bar{\phi}_t(x) - \nabla\bar{\phi}_t(y)\| \le \frac{1}{W}\sum_{i=0}^{w-1} u_i\,\|\nabla\phi_{t-i}(x) - \nabla\phi_{t-i}(y)\| \le L_\phi\,\|x-y\|.$$

Hence $\bar{\phi}_t$ is L_ϕ-smooth with the same constant.

By the descent lemma for smooth functions,

$$\bar{\phi}_t(x_{t+1}) \le \bar{\phi}_t(x_t) + \big\langle\nabla\bar{\phi}_t(x_t), x_{t+1}-x_t\big\rangle + \frac{L_\phi}{2}\|x_{t+1}-x_t\|^2.$$

Plug $x_{t+1} - x_t = -\eta_t^x\,\bar{q}_t^x$:

$$\bar{\phi}_t(x_{t+1}) \le \bar{\phi}_t(x_t) - \eta_t^x\langle\nabla\bar{\phi}_t(x_t), \bar{q}_t^x\rangle + \frac{L_\phi(\eta_t^x)^2}{2}\|\bar{q}_t^x\|^2.$$

Use the identity $-\langle a,b\rangle = -\frac{1}{2}\|a\|^2 - \frac{1}{2}\|b\|^2 + \frac{1}{2}\|a-b\|^2$ with $a = \nabla\bar{\phi}_t(x_t)$ and $b = \bar{q}_t^x$:

$$\bar{\phi}_t(x_{t+1}) \le \bar{\phi}_t(x_t) - \frac{\eta_t^x}{2}\|\nabla\bar{\phi}_t(x_t)\|^2 - \frac{\eta_t^x}{2}\|\bar{q}_t^x\|^2 + \frac{\eta_t^x}{2}\|\nabla\bar{\phi}_t(x_t) - \bar{q}_t^x\|^2 + \frac{L_\phi(\eta_t^x)^2}{2}\|\bar{q}_t^x\|^2.$$

Group the $\|\bar{q}_t^x\|^2$ terms:

$$-\frac{\eta_t^x}{2}+\frac{L_\phi(\eta_t^x)^2}{2}=-\frac{\eta_t^x}{2}\big(1-L_\phi\eta_t^x\big)\le-\frac{\eta_t^x}{4}\qquad\text{since }\ \eta_t^x\le\frac{1}{2L_\phi}.$$

Therefore,

$$\bar{\phi}_t(x_{t+1})\le\bar{\phi}_t(x_t)-\frac{\eta_t^x}{2}\|\nabla\bar{\phi}_t(x_t)\|^2-\frac{\eta_t^x}{4}\|\bar{q}_t^x\|^2+\frac{\eta_t^x}{2}\|\nabla\bar{\phi}_t(x_t)-\bar{q}_t^x\|^2.$$

Since $x_{t+1}-x_t=-\eta_t^x\bar{q}_t^x$, we have $\|\bar{q}_t^x\|^2=\|x_{t+1}-x_t\|^2/(\eta_t^x)^2$. Substitute this to get

$$\bar{\phi}_t(x_{t+1})\le\bar{\phi}_t(x_t)-\frac{\eta_t^x}{2}\|\nabla\bar{\phi}_t(x_t)\|^2-\frac{1}{4\,\eta_t^x}\|x_{t+1}-x_t\|^2+\frac{\eta_t^x}{2}\|\nabla\bar{\phi}_t(x_t)-\bar{q}_t^x\|^2,$$

which is exactly equation 42. $\square$

Lemma D.3 (Outer-direction mismatch under windowed averaging)**.** *Let $\bar{q}_t^x:=\bar{\nabla}_xf_t(x_t,y_t)+\lambda_t\big(\bar{\nabla}_xg_t(x_t,y_t)-\bar{\nabla}_xg_t(x_t,z_t)\big)$ with window weights $\{u_i\}_{i=0}^{w-1}$, $u_i\ge0$, $W=\sum_{i=0}^{w-1}u_i>0$, and define*

$$\nabla\bar{L}^\star_{\lambda_t,t}(x):=\frac{1}{W}\sum_{i=0}^{w-1}u_i\Big(\nabla_xf_{t-i}\big(x,\,y^\star_{\lambda,t-i}(x)\big)+\lambda_t\,\nabla_xg_{t-i}\big(x,\,y^\star_{\lambda,t-i}(x)\big)\Big).$$

Assume $\|\nabla_xf_s(x,y)-\nabla_xf_s(x,y')\|\le L_f\|y-y'\|$ and $\|\nabla_xg_s(x,y)-\nabla_xg_s(x,y')\|\le L_g\|y-y'\|$ for all s. Let $e_t^y:=\|y_t-y^\star_{\lambda,t}(x_t)\|$ and $e_t^z:=\|z_t-y_t^\star(x_t)\|$, and define the windowed penalized drift at x_t:

$$D_t^{(\lambda,u)}\ :=\ \frac{1}{W}\sum_{i=1}^{w-1}u_i\,\big\|y^\star_{\lambda,t-i}(x_t)-y^\star_{\lambda,t}(x_t)\big\|^2.$$

Then

$$\big\|\nabla\bar{L}^\star_{\lambda_t,t}(x_t)-\bar{q}_t^x\big\|^2\ \le\ 3(L_f+\lambda_tL_g)^2(e_t^y)^2\ +\ 3(\lambda_tL_g)^2(e_t^z)^2\ +\ 6(L_f+\lambda_tL_g)^2\,D_t^{(\lambda,u)}.\tag{43}$$

Moreover, letting $\Gamma_k^{(\lambda)}:=\sup_x\|y^\star_{\lambda,k+1}(x)-y^\star_{\lambda,k}(x)\|^2$ and $\Xi_u:=\frac{1}{W}\sum_{i=1}^{w-1}u_i\,i$, we have

$$D_t^{(\lambda,u)}\ \le\ \Xi_u\sum_{k=t-w}^{t-1}\Gamma_k^{(\lambda)}\qquad(\textit{with the convention }\Gamma_k^{(\lambda)}=0\textit{ for }k<1).\tag{44}$$

Proof. Write, for $s:=t-i$,

$$\nabla\bar{L}^\star_{\lambda_t,t}(x_t)-\bar{q}_t^x=\frac{1}{W}\sum_{i=0}^{w-1}u_i\Big\{\underbrace{\big[\nabla_xf_s(x_t,y^\star_{\lambda,s})-\nabla_xf_s(x_t,y_t)\big]}_{\mathcal{A}_s}+\lambda_t\underbrace{\big[\nabla_xg_s(x_t,y^\star_{\lambda,s})-\nabla_xg_s(x_t,y_t)\big]}_{\mathcal{B}_s}+\lambda_t\underbrace{\nabla_xg_s(x_t,z_t)}_{\mathcal{C}_s}\Big\}.$$

Add and subtract $\nabla_xg_s(x_t,y_t^\star(x_t))$ inside $\mathcal{C}_s$ and regroup:

$$\mathcal{C}_s=\big(\nabla_xg_s(x_t,z_t)-\nabla_xg_s(x_t,y_t^\star)\big)+\big(\nabla_xg_s(x_t,y_t^\star)-\nabla_xg_s(x_t,y^\star_{\lambda,t})\big)+\big(\nabla_xg_s(x_t,y^\star_{\lambda,t})-\nabla_xg_s(x_t,y^\star_{\lambda,s})\big).$$

Hence the average decomposes into three parts:

$$\frac{1}{W}\sum u_i(\mathcal{A}_s+\lambda_t\mathcal{B}_s)\quad\text{(depends on }y_t)\quad+\quad\lambda_t\frac{1}{W}\sum u_i\big(\nabla_xg_s(x_t,z_t)-\nabla_xg_s(x_t,y_t^\star)\big)\quad\text{(depends on }z_t)$$

$$+\quad\lambda_t\frac{1}{W}\sum u_i\big(\nabla_xg_s(x_t,y^\star_{\lambda,t})-\nabla_xg_s(x_t,y^\star_{\lambda,s})\big)\quad\text{(window drift).}$$

By the y-Lipschitz of ∇_xf_s,∇_xg_s,

$$\Big\|\frac{1}{W}\sum u_i(\mathcal{A}_s+\lambda_t\mathcal{B}_s)\Big\|\ \le\ (L_f+\lambda_tL_g)\,e_t^y,\qquad\Big\|\lambda_t\frac{1}{W}\sum u_i\big(\nabla_xg_s(x_t,z_t)-\nabla_xg_s(x_t,y_t^\star)\big)\Big\|\ \le\ \lambda_tL_g\,e_t^z.$$

For the drift term, again by Lipschitz in y,

$$\Big\|\lambda_t \frac{1}{W}\sum_{i=0}^{w-1} u_i\big(\nabla_x g_s(x_t, y^\star_{\lambda,t}) - \nabla_x g_s(x_t, y^\star_{\lambda,s})\big)\Big\| \;\le\; \lambda_t L_g \cdot \frac{1}{W}\sum_{i=1}^{w-1} u_i \left\|y^\star_{\lambda,t}(x_t) - y^\star_{\lambda,t-i}(x_t)\right\|.$$

Combine the three pieces and apply $(a+b+c)^2 \le 3a^2 + 3b^2 + 3c^2$ to obtain equation 43 with

$$D_t^{(\lambda,u)} = \frac{1}{W}\sum_{i=1}^{w-1} u_i \left\|y^\star_{\lambda,t-i}(x_t) - y^\star_{\lambda,t}(x_t)\right\|^2.$$

To prove equation 44, note $\|y^\star_{\lambda,t-i} - y^\star_{\lambda,t}\| \le \sum_{k=t-i}^{t-1}\|y^\star_{\lambda,k+1} - y^\star_{\lambda,k}\|$ so by Cauchy–Schwarz $\|y^\star_{\lambda,t-i} - y^\star_{\lambda,t}\|^2 \le i\sum_{k=t-i}^{t-1}\Gamma_k^{(\lambda)}$. Averaging with weights u_i/W gives

$$D_t^{(\lambda,u)} \le \frac{1}{W}\sum_{i=1}^{w-1} u_i\, i \sum_{k=t-i}^{t-1}\Gamma_k^{(\lambda)} \le \Big(\frac{1}{W}\sum_{i=1}^{w-1} u_i\, i\Big)\sum_{k=t-w}^{t-1}\Gamma_k^{(\lambda)} = \Xi_u \sum_{k=t-w}^{t-1}\Gamma_k^{(\lambda)}.$$

□

D.2 WINDOWED DETERMINISTIC ANALYSIS USING ϕ_t-SMOOTHNESS ONLY.

Recall the windowed operators and regret from Sec. 3:

$$\bar\nabla_u h_t(\cdot) := \frac{1}{W}\sum_{i=0}^{w-1} u_i\,\nabla_u h_{t-i}(\cdot) \quad (u\in\{x,y\},\ h\in\{f,g\}), \quad \mathsf{Reg}_T^{(u)} := \frac{1}{T}\sum_{t=1}^{T}\Big\|\frac{1}{W}\sum_{i=0}^{w-1} u_i\,\nabla\phi_{t-i}(x_t)\Big\|^2.$$

Algorithm 1 uses these windowed partials in all three updates.[2] Define the averaged outer objective $\bar\phi_t(x) := \frac{1}{W}\sum_{i=0}^{w-1} u_i\,\phi_{t-i}(x)$ and the averaged penalized proxy $\bar L^*_{\lambda,t}(x) := \frac{1}{W}\sum_{i=0}^{w-1} u_i\, L^*_{\lambda,t-i}(x)$.

Theorem D.1 (Deterministic WF^{2}OBO with windowed regret via ϕ–smoothness)**.** *Assume Assumption 1 and $\lambda \ge 2L_f/\mu$. Let L_ϕ be the uniform gradient Lipschitz constant in Lemma C.9. Choose stepsizes*

$$\eta_t^y \le \frac{1}{2(L_f + \lambda L_g)}, \qquad \eta_t^z \le \frac{1}{2L_g}, \qquad \eta_t^x \le \frac{1}{2L_\phi},$$

and run Algorithm 1 with window weights $\{u_i\}_{i=0}^{w-1}$ (sum W). Then, with $e_t^y := \|y_t - y^\star_{\lambda,t}(x_t)\|$, $e_t^z := \|z_t - y_t^\star(x_t)\|$, $\delta_t := (e_t^y)^2 + (e_t^z)^2$, there exist absolute constants C, C', C'', C_0 polynomial in $(\mu^{-1}, L_f, L_g, \rho_g)$ such that

$$\boxed{\mathsf{Reg}_T^{(u)} \;\le\; \frac{C}{\eta^x T}\Big(\bar\phi_1(x_1) - \inf_x \bar\phi_T(x)\Big) \;+\; \frac{C'}{T}\big(V_T^f + H_{2,T}\big) \;+\; \frac{C''}{W} \;+\; O\Big(\tfrac{1}{\lambda^2}\Big) \;+\; \frac{C_0\,\delta_0}{T}}.$$

In particular, with $u_i \equiv 1$ and $W = w = \lceil\sqrt{T}\rceil$, constant $\eta^x = \Theta(T^{-1/2})$, and any fixed $\lambda \ge 2L_f/\mu$,

$$\mathsf{Reg}_T^{(u)} = \tilde O(T^{-1/2}) + \tilde O\big((V_T^f + H_{2,T})/T\big) + O(T^{-1/2}) + O(\lambda^{-2}).$$

Proof. Using the L_ϕ–smoothness of $\bar\phi_t$ and the update $x_{t+1} = x_t - \eta_t^x \bar q_t^x$ with

$$\bar q_t^x := \bar\nabla_x f_t(x_t, y_t) + \lambda_t\big(\bar\nabla_x g_t(x_t, y_t) - \bar\nabla_x g_t(x_t, z_t)\big),$$

Lemma C.9 (applied to $\bar\phi_t$) gives, for $\eta_t^x \le 1/(2L_\phi)$,

$$\bar\phi_t(x_{t+1}) \le \bar\phi_t(x_t) - \frac{\eta_t^x}{2}\|\nabla\bar\phi_t(x_t)\|^2 - \frac{1}{4\eta_t^x}\|x_{t+1} - x_t\|^2 + \frac{\eta_t^x}{2}\left\|\nabla\bar\phi_t(x_t) - \bar q_t^x\right\|^2. \tag{45}$$

[2] Lines 2–4 of Alg. 1 in Sec. 3.

Step 2 (Bounding the outer-direction mismatch). Decompose, as in your Eq. (21)–(22) but averaged over the window,

$$\left\|\nabla\bar{\phi}_t(x_t)-\bar{q}_t^x\right\|^2 \;\le\; 2\left\|\nabla\bar{\phi}_t(x_t)-\nabla\bar{L}^\star_{\lambda,t}(x_t)\right\|^2+2\left\|\nabla\bar{L}^\star_{\lambda,t}(x_t)-\bar{q}_t^x\right\|^2.$$

By Lemma C.5 (gradient gap proxy vs. true) and Jensen,

$$\left\|\nabla\bar{\phi}_t(x_t)-\nabla\bar{L}^\star_{\lambda_t,t}(x_t)\right\|\le\frac{D_1}{\lambda_t},\qquad D_1=O(\ell\kappa^3).$$

For the second term, use the same Lipschitz argument as Eq. (22) on each window element and average:

$$\left\|\nabla\bar{L}^\star_{\lambda_t,t}(x_t)-\bar{q}_t^x\right\|^2\le 2(L_f+\lambda_tL_g)^2(e_t^y)^2+2(\lambda_tL_g)^2(e_t^z)^2.$$

Hence

$$\left\|\nabla\bar{\phi}_t(x_t)-\bar{q}_t^x\right\|^2\le\frac{2D_1^2}{\lambda_t^2}+4(L_f+\lambda_tL_g)^2(e_t^y)^2+4(\lambda_tL_g)^2(e_t^z)^2. \tag{46}$$

Step 3 (Summation and window telescoping). Sum equation 45 over $t=1,\dots,T$ and insert $\pm\bar{\phi}_{t-1}(x_t)$:

$$\begin{aligned}\frac12\sum_{t=1}^T\eta_t^x\|\nabla\bar{\phi}_t(x_t)\|^2+\frac14\sum_{t=1}^T\frac{\|x_{t+1}-x_t\|^2}{\eta_t^x}&\le\bar{\phi}_1(x_1)-\bar{\phi}_T(x_{T+1})+\sum_{t=2}^T\big(\bar{\phi}_t(x_t)-\bar{\phi}_{t-1}(x_t)\big)\\&\quad+\frac12\sum_{t=1}^T\eta_t^x\left\|\nabla\bar{\phi}_t(x_t)-\bar{q}_t^x\right\|^2.\end{aligned}\tag{47}$$

By definition of the average,

$$\begin{aligned}&\sum_{t=2}^T\big(\bar{\phi}_t(x_t)-\bar{\phi}_{t-1}(x_t)\big)\\&\qquad\le\sum_{t=2}^T|f_t(x_t,y_t^*(x_t))-f_{t-1}(x_t,y_t^*(x_t))|+L_f\sum_{t=2}^T\|y_t^*(x_t)-y_{t-1}^*(x_t)\|\\&\qquad\le V_T^f+L_fH_{1,T},\end{aligned}\tag{48}$$

where we used the definition of V_T^f.[3]

Insert equation 46 into equation 47 to obtain

$$\begin{aligned}\frac12\sum_{t=1}^T\eta_t^x\|\nabla\bar{\phi}_t(x_t)\|^2+\frac14\sum_{t=1}^T\frac{\|x_{t+1}-x_t\|^2}{\eta_t^x}&\le\bar{\phi}_1(x_1)-\inf_x\bar{\phi}_T(x)+V_T^f+L_fH_{1,T}\\&\quad+\sum_{t=1}^T\eta_t^x\Big[(L_f+\lambda_tL_g)^2(e_t^y)^2+(\lambda_tL_g)^2(e_t^z)^2\Big]+O\Big(\tfrac{1}{\lambda_t^2}\Big).\end{aligned}\tag{49}$$

Step 4 (Inner tracking recursion & absorption). As in Lemma D.1, with $\eta_t^y\le 1/(L_f+\lambda L_g)$ and $\eta_t^z\le 1/L_g$, the inner errors obey the linear recursion (we restate it for completeness):

$$\delta_{t+1}\;\le\;(1-c_t)\,\delta_t\;+\;C_x\|x_{t+1}-x_t\|^2\;+\;K_y\,\Gamma_t^{(\lambda)}\;+\;K_z\,\Gamma_t\;+\;K_y\Big[\tfrac{C_f^2}{\lambda_t^2\mu^2}+\tfrac{C_f^2}{\lambda_{t+1}^2\mu^2}\Big],\tag{50}$$

with $c_t=\Theta(\lambda\mu\eta_t^y)+\Theta(\mu\eta_t^z)$, $C_x=\Theta(L_g^2\mu^{-2})(K_y+K_z)$, and $\Gamma_t:=\sup_x\|y_t^\star(x)-y_{t-1}^\star(x)\|^2$, $\Gamma_t^{(\lambda)}:=\sup_x\|y^\star_{\lambda,t}(x)-y^\star_{\lambda,t-1}(x)\|^2$. This proof uses (i) strong convexity/smoothness in y (Lemma C.2), (ii) Lipschitzness of $y^\star(\cdot)$ in x (Lemma C.1), and (iii) the same weighted Young/telescoping manipulations as Equations (28)–(33). The only change under windowing is that the y,z steps take the gradient of the averaged functions; this yields the same contraction towards the minimizer of the

[3]Precisely: drop at most w boundary terms and divide by W; the $O(1/\lambda)$ proxy-to-true differences aggregate into the $O(1/\lambda^2)$ term collected later.

averaged lower problems and an additional drift term that is bounded by the *same* one-step path lengths $\Gamma_t, \Gamma_t^{(\lambda)}$ (averaging creates convex combinations of the same differences). Thus equation 50 holds up to constants.

Summing equation 50 over t and using $\sum_t \delta_t$–to–$\sum_t \|x_{t+1} - x_t\|^2$ absorption as in your Theorem 4.1 proof (choose η_t^x small enough so that $\eta_t^x (L_f + \lambda L_g)^2 \delta_t$ is absorbed by $\sum \frac{\|x_{t+1} - x_t\|^2}{\eta_t^x}$ and the contraction mass $\sum c_t \delta_t$), we obtain

$$\sum_{t=1}^{T} \eta_t^x \Big[(L_f + \lambda L_g)^2 (e_t^y)^2 + (\lambda L_g)^2 (e_t^z)^2 \Big] \;\le\; \tilde{C}\,\delta_0 \;+\; \tilde{C} \sum_{t=1}^{T} \frac{\|x_{t+1} - x_t\|^2}{\eta_t^x} \;+\; \tilde{C} \sum_{t=1}^{T} (\Gamma_t^{(\lambda)} + \Gamma_t) \;+\; O\Big(\tfrac{1}{\lambda^2}\Big). \quad (51)$$

Plug equation 51 into equation 49 and absorb the $\sum \frac{\|x_{t+1}-x_t\|^2}{\eta_t^x}$ term to the LHS (by taking η_t^x small, e.g., fixed $\eta^x \le 1/(2L_\phi)$). This yields

$$\frac{1}{2} \sum_{t=1}^{T} \eta_t^x \|\nabla \bar{\phi}_t(x_t)\|^2 \le \bar{\phi}_1(x_1) - \inf_x \bar{\phi}_T(x) + V_T^f + O(1/W) + C_0 \delta_0 + C_1 \sum_{t=1}^{T} (\Gamma_t^{(\lambda)} + \Gamma_t) + O\Big(\tfrac{1}{\lambda^2}\Big). \quad (52)$$

Step 5 (From proxies to f-budgets only). By Lemma C.3 and Lemma C.4,

$$\Gamma_t^{(\lambda)} \;\le\; \Gamma_t + O(1/\lambda^2), \qquad |\phi_t - \phi_{t-1}| \;\le\; |f_t - f_{t-1}| + O(1/\lambda),$$

so $\sum_t \Gamma_t^{(\lambda)} \le H_{2,T} + O(T/\lambda^2)$ and the value drift reduces to V_T^f up to $O(T/\lambda)$. Collecting these into the existing $O(1/\lambda^2)$ term, equation 52 becomes

$$\frac{1}{2} \sum_{t=1}^{T} \eta_t^x \|\nabla \bar{\phi}_t(x_t)\|^2 \;\le\; \bar{\phi}_1(x_1) - \inf_x \bar{\phi}_T(x) \;+\; V_T^f \;+\; O(1/W) \;+\; C_0 \delta_0 \;+\; C_2 H_{2,T} \;+\; O\Big(\tfrac{1}{\lambda^2}\Big).$$

Step 6 (Relating $\nabla \bar{\phi}_t$ to windowed regret). By definition,

$$\nabla \bar{\phi}_t(x_t) = \frac{1}{W} \sum_{i=0}^{w-1} u_i \, \nabla \phi_{t-i}(x_t),$$

so the LHS is exactly $\sum_t \eta_t^x \big\| \frac{1}{W} \sum_i u_i \nabla \phi_{t-i}(x_t) \big\|^2$. Dividing both sides by $\sum_{t=1}^{T} \eta_t^x \ge T \min_t \eta_t^x$ yields the claimed bound on $\mathsf{Reg}_T^{(u)}$ with initial gap $\bar{\phi}_1(x_1) - \inf_x \bar{\phi}_T(x)$, budgets V_T^f and $H_{2,T}$, the window edge term $O(1/W)$, the proxy term $O(1/\lambda^2)$, and the initial inner error $C_0 \delta_0 / T$. $\square$

E PROOFS FOR STOCHASTIC ANALYSIS

We work with descent of the penalized proxy $L^*_{\lambda,t}$, for which the following inexact descent lemma is convenient.

Lemma E.1. *Assume that $L^*_{\lambda,t}$ is L-smooth which derived under Assumption 1 in Lemma C.8. If $\eta_t^x \le \frac{1}{2L}$ and we set*

$$\mathsf{E}_t^x \;:=\; d_t^x - \nabla L^*_{\lambda,t}(x_t), \qquad \Delta x_t \;:=\; x_{t+1} - x_t \;=\; -\,\eta_t^x \, d_t^x,$$

then

$$L^*_{\lambda,t}(x_{t+1}) \;\le\; L^*_{\lambda,t}(x_t) - \frac{\eta_t^x}{2} \big\|\nabla L^*_{\lambda,t}(x_t)\big\|^2 - \frac{1}{4\eta_t^x} \|\Delta x_t\|^2 + \frac{\eta_t^x}{2} \|\mathsf{E}_t^x\|^2.$$

Proof. Write $g_t := \nabla L^*_{\lambda,t}(x_t)$ for brevity. By L-smoothness,

$$L^*_{\lambda,t}(x_{t+1}) \;\le\; L^*_{\lambda,t}(x_t) + \langle g_t, \Delta x_t \rangle + \frac{L}{2} \|\Delta x_t\|^2. \quad (53)$$

We will upper-bound the inner product $\langle g_t, \Delta x_t \rangle$ via the following exact algebraic identity, valid for any vectors a, b and any $\eta > 0$:

$$\langle a, b \rangle \;=\; -\frac{1}{2\eta} \|b\|^2 \;-\; \frac{\eta}{2} \|a\|^2 \;+\; \frac{1}{2\eta} \|\, b + \eta a \,\|^2. \quad (54)$$

(Indeed, expanding the last norm gives $\frac{1}{2\eta}(\|b\|^2 + 2\eta\langle a, b\rangle + \eta^2\|a\|^2) = \langle a, b\rangle + \frac{1}{2\eta}\|b\|^2 + \frac{\eta}{2}\|a\|^2$, which rearranges to equation 54.)

Apply equation 54 with $a = g_t$, $b = \Delta x_t$ and $\eta = \eta_t^x$:

$$\langle g_t, \Delta x_t\rangle \;\le\; -\frac{1}{2\eta_t^x}\|\Delta x_t\|^2 \;-\; \frac{\eta_t^x}{2}\|g_t\|^2 \;+\; \frac{1}{2\eta_t^x}\|\,\Delta x_t + \eta_t^x g_t\,\|^2.$$

Plug this into equation 53 to obtain

$$L_{\lambda,t}^*(x_{t+1}) \;\le\; L_{\lambda,t}^*(x_t) - \frac{\eta_t^x}{2}\|g_t\|^2 - \Big(\frac{1}{2\eta_t^x} - \frac{L}{2}\Big)\|\Delta x_t\|^2 + \frac{1}{2\eta_t^x}\|\,\Delta x_t + \eta_t^x g_t\,\|^2. \tag{55}$$

Next, use the update $\Delta x_t = -\eta_t^x d_t^x$ with $d_t^x = g_t + \mathsf{E}_t^x$:

$$\Delta x_t + \eta_t^x g_t \;=\; -\eta_t^x(g_t + \mathsf{E}_t^x) + \eta_t^x g_t \;=\; -\,\eta_t^x\,\mathsf{E}_t^x,$$

so the last term in equation 55 becomes

$$\frac{1}{2\eta_t^x}\|\,\Delta x_t + \eta_t^x g_t\,\|^2 \;=\; \frac{1}{2\eta_t^x}\,\|\eta_t^x \mathsf{E}_t^x\|^2 \;=\; \frac{\eta_t^x}{2}\,\|\mathsf{E}_t^x\|^2.$$

Finally, enforce $\eta_t^x \le \frac{1}{2L}$ so that

$$\frac{1}{2\eta_t^x} - \frac{L}{2} \;\ge\; \frac{1}{2\eta_t^x} - \frac{1}{4\eta_t^x} \;=\; \frac{1}{4\eta_t^x},$$

and equation 55 yields

$$L_{\lambda,t}^*(x_{t+1}) \;\le\; L_{\lambda,t}^*(x_t) - \frac{\eta_t^x}{2}\|g_t\|^2 - \frac{1}{4\eta_t^x}\,\|\Delta x_t\|^2 + \frac{\eta_t^x}{2}\,\|\mathsf{E}_t^x\|^2,$$

which is exactly the claimed inequality. $\square$

Lemma E.2 (Variance-reduced MSE recursion)**.** *Let d_t^z, d_t^y, d_t^x be the control-variates in equation 12. Define the estimator errors $\epsilon_t^u := d_t^u - G_t^u$ for $u \in \{x, y, z\}$. Under Assumption 3, conditioning on $\mathcal{F}_t$,*

$$\begin{aligned}
\mathbb{E}\big[\|\epsilon_t^z\|^2 \mid \mathcal{F}_t\big] &\le \Big(1 - \tfrac{\gamma_t^z}{2}\Big)\|\epsilon_{t-1}^z\|^2 + \Big(1 - \tfrac{\gamma_t^z}{2}\Big)\Big(1 + \tfrac{2}{\gamma_t^z}\Big)\cdot\|\nabla_y g_t(x_{t-1}, y_{t-1}) - \nabla_y g_{t-1}(x_{t-1}, y_{t-1})\| \\
&\quad + \frac{2\,\overline{L}_{g,y}^2}{|B_t^{gy}|}\Big(\|x_t - x_{t-1}\|^2 + \|z_t - z_{t-1}\|^2\Big) + \frac{2(\gamma_t^z)^2}{|B_t^{gy}|}\,\sigma_{g,y}^2
\end{aligned} \tag{56}$$

$$\begin{aligned}
\mathbb{E}\big[\|\epsilon_t^y\|^2 \mid \mathcal{F}_t\big] &\le \Big(1 - \tfrac{\gamma_t^y}{2}\Big)\|\epsilon_{t-1}^y\|^2 \\
&\quad + \Big(1 - \tfrac{\gamma_t^y}{2}\Big)\Big(1 + \tfrac{2}{\gamma_t^y}\Big)\|\big(\nabla_y f_{t-1} + \lambda_{t-1}\nabla_y g_{t-1}\big)(x_{t-1}, y_{t-1}) - \big(\nabla_y f_t + \lambda_t \nabla_y g_t\big)(x_{t-1}, y_{t-1})\|^2 \\
&\quad + \left[\frac{2\,\overline{L}_{fy}^2}{|B_t^{fy}|} + \frac{2\,\lambda_t^2\,\overline{L}_{gy}^2}{|B_t^{gy}|}\right]\big(\|x_t - x_{t-1}\|^2 + \|y_t - y_{t-1}\|^2\big) \\
&\quad + \frac{2(\gamma_t^y)^2}{|B_t^{fy}|}\sigma_{f,y}^2 + \frac{2\lambda_t^2(\gamma_t^y)^2}{|B_t^{gy}|}\sigma_{g,y}^2.
\end{aligned} \tag{57}$$

$$\begin{aligned}
\mathbb{E}\big[\|\epsilon_t^x\|^2 \mid \mathcal{F}_t\big] &\le \Big(1 - \tfrac{\gamma_t^x}{2}\Big)\|\epsilon_{t-1}^x\|^2 + \Big(1 - \tfrac{\gamma_t^x}{2}\Big)\Big(1 + \tfrac{2}{\gamma_t^x}\Big)\|\Delta_{t,x}^{L_{\lambda_t,t}}\|^2 \\
&\quad + \left[\frac{2\,\overline{L}_{fx}^2}{|B_t^{fx}|} + \frac{4\,\lambda_t^2\,\overline{L}_{gx}^2}{|B_t^{gx}|}\right]\cdot\big(\|x_t - x_{t-1}\|^2 + \|y_t - y_{t-1}\|^2\big) + \frac{4\,\lambda_t^2\,\overline{L}_{gx}^2}{|B_t^{gx}|}\cdot\|z_t - z_{t-1}\|^2 \\
&\quad + \frac{2(\gamma_t^x)^2}{|B_t^{fx}|}\,\sigma_{f,x}^2 + \frac{4\lambda_t^2(\gamma_t^x)^2}{|B_t^{gx}|}\,\sigma_{g,x}^2.
\end{aligned} \tag{58}$$

Proof. Throughout, the batches are (i) reused *within* the difference in equation 12 and (ii) independent across t and of $\mathcal{F}_t$.

Case z. We start from the STORM-style update for the z-control variate (the batch B_t^{gy} is *reused within the difference*):

$$d_t^z = \nabla_y g_t(x_t, z_t; B_t^{gy}) + (1-\gamma_t^z)\Big(d_{t-1}^z - \nabla_y g_t(x_{t-1}, z_{t-1}; B_t^{gy})\Big).$$

Let the population gradient and the estimation error be

$$G_t^z := \nabla_y g_t(x_t, z_t), \qquad \epsilon_t^z := d_t^z - G_t^z,$$

and define the "shifted" previous error (against the round-t function)

$$\epsilon_{t-1}^{z,(t)} := d_{t-1}^z - \nabla_y g_t(x_{t-1}, z_{t-1}).$$

Adding and subtracting population terms only inside the reuse bracket yields the clean decomposition

$$\epsilon_t^z = (1-\gamma_t^z)\,\epsilon_{t-1}^{z,(t)} + \xi_t^z, \qquad \xi_t^z := \big(\nabla_y g_t(x_t, z_t; B_t^{gy}) - \nabla_y g_t(x_t, z_t)\big) - (1-\gamma_t^z)\big(\nabla_y g_t(x_{t-1}, z_{t-1}; B_t^{gy}) - \nabla_y g_t(x_{t-1}, z_{t-1})\big). \tag{59}$$

By unbiasedness and independence of B_t^{gy} from $\mathcal{F}_t$ (Assumption 3), $\mathbb{E}[\xi_t^z \mid \mathcal{F}_t] = 0$. Conditioning on $\mathcal{F}_t$ and using the vanishing cross term,

$$\mathbb{E}\big[\|\epsilon_t^z\|^2 \mid \mathcal{F}_t\big] = (1-\gamma_t^z)^2 \|\epsilon_{t-1}^{z,(t)}\|^2 + \mathbb{E}\big[\|\xi_t^z\|^2 \mid \mathcal{F}_t\big]. \tag{60}$$

Rewrite ξ_t^z as

$$\begin{aligned}\xi_t^z &= \underbrace{\big(\nabla_y g_t(x_t, z_t; B_t^{gy}) - \nabla_y g_t(x_{t-1}, z_{t-1}; B_t^{gy})\big) - \big(\nabla_y g_t(x_t, z_t) - \nabla_y g_t(x_{t-1}, z_{t-1})\big)}_{=:\, v_t^{\text{diff}}} \\ &\quad + \gamma_t^z \underbrace{\big(\nabla_y g_t(x_{t-1}, z_{t-1}) - \nabla_y g_t(x_{t-1}, z_{t-1}; B_t^{gy})\big)}_{=:\, v_t^{\text{var}}}.\end{aligned} \tag{61}$$

By the expected-smoothness/variance bounds in Assumption 3 and the reuse of B,

$$\mathbb{E}\big[\|\xi_t^z\|^2 \mid \mathcal{F}_t\big] \le \frac{2\,\overline{L}_{g,y}^2}{|B_t^{gy}|}\Big(\|x_t - x_{t-1}\|^2 + \|z_t - z_{t-1}\|^2\Big) + \frac{2(\gamma_t^z)^2}{|B_t^{gy}|}\,\sigma_{g,y}^2. \tag{62}$$

We introduce the inter-round drift at the previous point

$$\Delta_{t,y}^g := \nabla_y g_{t-1}(x_{t-1}, z_{t-1}) - \nabla_y g_t(x_{t-1}, z_{t-1}),$$

so that $\epsilon_{t-1}^{z,(t)} = \epsilon_{t-1}^z + \Delta_{t,y}^g$ with $\epsilon_{t-1}^z := d_{t-1}^z - \nabla_y g_{t-1}(x_{t-1}, z_{t-1})$. By Young's inequality, for any $\theta_t \in (0,1]$,

$$\|\epsilon_{t-1}^{z,(t)}\|^2 \le (1+\theta_t)\,\|\epsilon_{t-1}^z\|^2 + \Big(1 + \tfrac{1}{\theta_t}\Big)\|\Delta_{t,y}^g\|^2. \tag{63}$$

Plugging equation 62 and equation 63 into equation 60 with the convenient choice $\theta_t = \gamma_t^z/2$ and using $(1-\gamma)^2(1+\gamma/2) \le 1-\gamma/2$ for $\gamma \in (0,1]$, we obtain

$$\mathbb{E}\big[\|\epsilon_t^z\|^2 \mid \mathcal{F}_t\big] \le \Big(1 - \tfrac{\gamma_t^z}{2}\Big)\|\epsilon_{t-1}^z\|^2 + \Big(1 - \tfrac{\gamma_t^z}{2}\Big)\Big(1 + \tfrac{2}{\gamma_t^z}\Big)\|\Delta_{t,y}^g\|^2 + \frac{2\,\overline{L}_{g,y}^2}{|B_t^{gy}|}\Big(\|x_t - x_{t-1}\|^2 + \|z_t - z_{t-1}\|^2\Big) + \frac{2(\gamma_t^z)^2}{|B_t^{gy}|}\,\sigma_{g,y}^2. \tag{64}$$

Case y. From the STORM-style update for the y-control variate,

$$d_t^y = \nabla_y f_t(x_t, y_t; B_t^{fy}) + \lambda_t \nabla_y g_t(x_t, y_t; B_t^{gy}) + (1-\gamma_t^y)\Big(d_{t-1}^y - \nabla_y f_t(x_{t-1}, y_{t-1}; B_t^{fy}) - \lambda_{t-1}\nabla_y g_t(x_{t-1}, y_{t-1}; B_t^{gy})\Big).$$

Let

$$G_t^y := \nabla_y f_t(x_t, y_t) + \lambda_t \nabla_y g_t(x_t, y_t), \qquad \epsilon_t^y := d_t^y - G_t^y,$$

and define the *shifted* previous error (measured against round-t composites)

$$\epsilon_{t-1}^{y,(t)} := d_{t-1}^y - \nabla_y f_t(x_{t-1}, y_{t-1}) - \lambda_{t-1}\nabla_y g_t(x_{t-1}, y_{t-1}).$$

Add and subtract only population terms *inside* the reuse bracket to obtain

$$\epsilon_t^y = (1-\gamma_t^y)\,\epsilon_{t-1}^{y,(t)} + \xi_t^y, \tag{65}$$

where

$$\xi_t^y := \underbrace{\big(\nabla_y f_t(x_t,y_t;B_t^{fy}) - \nabla_y f_t(x_t,y_t)\big) - (1-\gamma_t^y)\big(\nabla_y f_t(x_{t-1},y_{t-1};B_t^{fy}) - \nabla_y f_t(x_{t-1},y_{t-1})\big)}_{=:\,\xi_{t,f}^y}$$
$$+\,\lambda_t\,\underbrace{\big(\nabla_y g_t(x_t,y_t;B_t^{gy}) - \nabla_y g_t(x_t,y_t)\big) - (1-\gamma_t^y)\big(\nabla_y g_t(x_{t-1},y_{t-1};B_t^{gy}) - \nabla_y g_t(x_{t-1},y_{t-1})\big)}_{=:\,\xi_{t,g}^y}.$$

By unbiasedness and independence of B_t^{fy}, B_t^{gy} from $\mathcal{F}_t$ (Assumption 3), we have $\mathbb{E}[\xi_t^y \mid \mathcal{F}_t] = 0$. Hence, conditioning on $\mathcal{F}_t$,

$$\mathbb{E}\big[\|\epsilon_t^y\|^2 \mid \mathcal{F}_t\big] = (1-\gamma_t^y)^2\|\epsilon_{t-1}^{y,(t)}\|^2 + \mathbb{E}\big[\|\xi_t^y\|^2 \mid \mathcal{F}_t\big]. \tag{66}$$

Bounding the noise term. We first handle the f-part. Rewrite

$$\xi_{t,f}^y = \underbrace{\big(\nabla_y f_t(x_t,y_t;B_t^{fy}) - \nabla_y f_t(x_{t-1},y_{t-1};B_t^{fy})\big) - \big(\nabla_y f_t(x_t,y_t) - \nabla_y f_t(x_{t-1},y_{t-1})\big)}_{=:\,\upsilon_{t,f,\mathrm{diff}}^y}$$
$$+\;\gamma_t^y\,\underbrace{\big(\nabla_y f_t(x_{t-1},y_{t-1}) - \nabla_y f_t(x_{t-1},y_{t-1};B_t^{fy})\big)}_{=:\,\upsilon_{t,f,\mathrm{var}}^y}. \tag{67}$$

By the expected-smoothness bound in Assumption 3 (applied with the same batch reused inside the difference),

$$\mathbb{E}\big[\|\upsilon_{t,f,\mathrm{diff}}^y\|^2 \mid \mathcal{F}_t\big] \;\le\; \frac{\overline{L}_{fy}^2}{|B_t^{fy}|}\big(\|x_t-x_{t-1}\|^2 + \|y_t-y_{t-1}\|^2\big), \qquad \mathbb{E}\big[\|\upsilon_{t,f,\mathrm{var}}^y\|^2 \mid \mathcal{F}_t\big] \;\le\; \frac{\sigma_{f,y}^2}{|B_t^{fy}|}.$$

Thus,

$$\mathbb{E}\big[\|\xi_{t,f}^y\|^2 \mid \mathcal{F}_t\big] \;\le\; \frac{2\,\overline{L}_{fy}^2}{|B_t^{fy}|}\big(\|x_t-x_{t-1}\|^2 + \|y_t-y_{t-1}\|^2\big) + \frac{2(\gamma_t^y)^2}{|B_t^{fy}|}\,\sigma_{f,y}^2. \tag{68}$$

The g-part is identical mutatis mutandis:

$$\xi_{t,g}^y = \underbrace{\big(\nabla_y g_t(x_t,y_t;B_t^{gy}) - \nabla_y g_t(x_{t-1},y_{t-1};B_t^{gy})\big) - \big(\nabla_y g_t(x_t,y_t) - \nabla_y g_t(x_{t-1},y_{t-1})\big)}_{=:\,\upsilon_{t,g,\mathrm{diff}}^y}$$
$$+\;\gamma_t^y\,\underbrace{\big(\nabla_y g_t(x_{t-1},y_{t-1}) - \nabla_y g_t(x_{t-1},y_{t-1};B_t^{gy})\big)}_{=:\,\upsilon_{t,g,\mathrm{var}}^y}, \tag{69}$$

and Assumption 3 yields

$$\mathbb{E}\big[\|\upsilon_{t,g,\mathrm{diff}}^y\|^2 \mid \mathcal{F}_t\big] \;\le\; \frac{\overline{L}_{gy}^2}{|B_t^{gy}|}\big(\|x_t-x_{t-1}\|^2 + \|y_t-y_{t-1}\|^2\big), \quad \mathbb{E}\big[\|\upsilon_{t,g,\mathrm{var}}^y\|^2 \mid \mathcal{F}_t\big] \;\le\; \frac{\sigma_{g,y}^2}{|B_t^{gy}|}.$$

Therefore,

$$\mathbb{E}\big[\|\xi_{t,g}^y\|^2 \mid \mathcal{F}_t\big] \;\le\; \frac{2\,\overline{L}_{gy}^2}{|B_t^{gy}|}\big(\|x_t-x_{t-1}\|^2 + \|y_t-y_{t-1}\|^2\big) + \frac{2(\gamma_t^y)^2}{|B_t^{gy}|}\,\sigma_{g,y}^2. \tag{70}$$

Using $\|u+\lambda v\|^2 \le 2\|u\|^2 + 2\lambda^2\|v\|^2$ and combining equation 68–equation 70,

$$\mathbb{E}\big[\|\xi_t^y\|^2 \mid \mathcal{F}_t\big] \;\le\; \frac{2\,\overline{L}_{fy}^2}{|B_t^{fy}|}\big(\|x_t-x_{t-1}\|^2 + \|y_t-y_{t-1}\|^2\big) + \frac{2\,\lambda_t^2\,\overline{L}_{gy}^2}{|B_t^{gy}|}\big(\|x_t-x_{t-1}\|^2 + \|y_t-y_{t-1}\|^2\big) + \frac{2(\gamma_t^y)^2}{|B_t^{fy}|}\sigma_{f,y}^2 + \frac{2\lambda_t^2(\gamma_t^y)^2}{|B_t^{gy}|}\sigma_{g,y}^2. \tag{71}$$

Define the inter-round drift at (x_{t-1}, y_{t-1}) (note the possible change in λ):

$$\Delta_{t,y}^{L_{\lambda_t,t}} := \big(\nabla_y f_{t-1} + \lambda_{t-1}\nabla_y g_{t-1}\big)(x_{t-1}, y_{t-1}) - \big(\nabla_y f_t + \lambda_{t-1}\nabla_y g_t\big)(x_{t-1}, y_{t-1}),$$

so that $\epsilon_{t-1}^{y,(t)} = \epsilon_{t-1}^{y} + \Delta_t^y$, with $\epsilon_{t-1}^y := d_{t-1}^y - \nabla_y f_{t-1}(x_{t-1}, y_{t-1}) - \lambda_{t-1}\nabla_y g_{t-1}(x_{t-1}, y_{t-1})$. By Young's inequality, for any $\theta_t \in (0, 1]$,

$$\|\epsilon_{t-1}^{y,(t)}\|^2 \le (1+\theta_t)\|\epsilon_{t-1}^y\|^2 + \Big(1 + \tfrac{1}{\theta_t}\Big)\left\|\Delta_{t,y}^{L_{\lambda_t,t}}\right\|^2. \tag{72}$$

Putting things together. Plug equation 71 and equation 72 into equation 66 with $\theta_t = \gamma_t^y/2$; using $(1-\gamma)^2(1+\gamma/2) \le 1-\gamma/2$ for $\gamma \in (0, 1]$, we obtain

$$\begin{aligned}
\mathbb{E}\big[\|\epsilon_t^y\|^2 \mid \mathcal{F}_t\big] \le{}& \Big(1 - \tfrac{\gamma_t^y}{2}\Big)\|\epsilon_{t-1}^y\|^2 \\
&+ \Big(1 - \tfrac{\gamma_t^y}{2}\Big)\Big(1 + \tfrac{2}{\gamma_t^y}\Big)\left\|\Delta_{t,y}^{L_{\lambda_t,t}}\right\|^2 \\
&+ \frac{2\,\overline{L}_{fy}^2}{|B_t^{fy}|}\big(\|x_t - x_{t-1}\|^2 + \|y_t - y_{t-1}\|^2\big) + \frac{2\,\lambda_t^2\,\overline{L}_{gy}^2}{|B_t^{gy}|}\big(\|x_t - x_{t-1}\|^2 + \|y_t - y_{t-1}\|^2\big) \\
&+ \frac{2(\gamma_t^y)^2}{|B_t^{fy}|}\sigma_{f,y}^2 + \frac{2\lambda_t^2(\gamma_t^y)^2}{|B_t^{gy}|}\sigma_{g,y}^2,
\end{aligned} \tag{73}$$

which is the desired recursion for y.

Case x. The STORM-style update for x reads

$$\begin{aligned}
d_t^x ={}& \nabla_x f_t(x_t, y_t; B_t^{fx}) + \lambda_t\Big(\nabla_x g_t(x_t, y_t; B_t^{gx}) - \nabla_x g_t(x_t, z_t; B_t^{gx})\Big) \\
&+ (1-\gamma_t^x)\Big(d_{t-1}^x - \nabla_x f_t(x_{t-1}, y_{t-1}; B_t^{fx}) - \lambda_{t-1}\big(\nabla_x g_t(x_{t-1}, y_{t-1}; B_t^{gx}) - \nabla_x g_t(x_{t-1}, z_{t-1}; B_t^{gx})\big)\Big).
\end{aligned}$$

Define

$$G_t^x := \nabla_x f_t(x_t, y_t) + \lambda_t\big(\nabla_x g_t(x_t, y_t) - \nabla_x g_t(x_t, z_t)\big), \qquad \epsilon_t^x := d_t^x - G_t^x,$$

and the shifted error

$$\epsilon_{t-1}^{x,(t)} := d_{t-1}^x - \nabla_x f_t(x_{t-1}, y_{t-1}) - \lambda_{t-1}\big(\nabla_x g_t(x_{t-1}, y_{t-1}) - \nabla_x g_t(x_{t-1}, z_{t-1})\big).$$

As before,

$$\epsilon_t^x = (1-\gamma_t^x)\,\epsilon_{t-1}^{x,(t)} + \xi_t^x, \tag{74}$$

with the (zero-mean) noise

$$\begin{aligned}
\xi_t^x :={}& \underbrace{\big(\nabla_x f_t(x_t, y_t; B_t^{fx}) - \nabla_x f_t(x_t, y_t)\big) - (1-\gamma_t^x)\big(\nabla_x f_t(x_{t-1}, y_{t-1}; B_t^{fx}) - \nabla_x f_t(x_{t-1}, y_{t-1})\big)}_{=:\,\xi_{t,f}^x} \\
&+ \lambda_t\underbrace{\Big[\big(\nabla_x g_t(x_t, y_t; B_t^{gx}) - \nabla_x g_t(x_t, y_t)\big) - (1-\gamma_t^x)\big(\nabla_x g_t(x_{t-1}, y_{t-1}; B_t^{gx}) - \nabla_x g_t(x_{t-1}, y_{t-1})\big)\Big]}_{=:\,\xi_{t,g}^{x,y}} \\
&- \lambda_t\underbrace{\Big[\big(\nabla_x g_t(x_t, z_t; B_t^{gx}) - \nabla_x g_t(x_t, z_t)\big) - (1-\gamma_t^x)\big(\nabla_x g_t(x_{t-1}, z_{t-1}; B_t^{gx}) - \nabla_x g_t(x_{t-1}, z_{t-1})\big)\Big]}_{=:\,\xi_{t,g}^{x,z}}.
\end{aligned}$$

Conditioning on $\mathcal{F}_t$ and using $\mathbb{E}[\xi_t^x \mid \mathcal{F}_t] = 0$,

$$\mathbb{E}\big[\|\epsilon_t^x\|^2 \mid \mathcal{F}_t\big] = (1-\gamma_t^x)^2\|\epsilon_{t-1}^{x,(t)}\|^2 + \mathbb{E}\big[\|\xi_t^x\|^2 \mid \mathcal{F}_t\big]. \tag{75}$$

Bounding the noise term. Proceed term-by-term as in the y-case. For the f-part,

$$\xi_{t,f}^x = \underbrace{\big(\nabla_x f_t(x_t, y_t; B_t^{fx}) - \nabla_x f_t(x_{t-1}, y_{t-1}; B_t^{fx})\big) - \big(\nabla_x f_t(x_t, y_t) - \nabla_x f_t(x_{t-1}, y_{t-1})\big)}_{=:\,\upsilon_{t,f,\mathrm{diff}}^x}$$

$$+\ \gamma_t^x \underbrace{\left(\nabla_x f_t(x_{t-1}, y_{t-1}) - \nabla_x f_t(x_{t-1}, y_{t-1}; B_t^{fx})\right)}_{=:\ \upsilon_{t,f,\mathrm{var}}^x}, \tag{76}$$

which implies

$$\mathbb{E}\big[\|\xi_{t,f}^x\|^2 \mid \mathcal{F}_t\big] \ \le\ \frac{2\,\overline{L}_{fx}^2}{|B_t^{fx}|}\big(\|x_t - x_{t-1}\|^2 + \|y_t - y_{t-1}\|^2\big) + \frac{2(\gamma_t^x)^2}{|B_t^{fx}|}\,\sigma_{f,x}^2. \tag{77}$$

For the two g-parts, similarly,

$$\mathbb{E}\big[\|\xi_{t,g}^{x,y}\|^2 \mid \mathcal{F}_t\big] \ \le\ \frac{2\,\overline{L}_{gx}^2}{|B_t^{gx}|}\big(\|x_t - x_{t-1}\|^2 + \|y_t - y_{t-1}\|^2\big) + \frac{2(\gamma_t^x)^2}{|B_t^{gx}|}\,\sigma_{g,x}^2,$$

$$\mathbb{E}\big[\|\xi_{t,g}^{x,z}\|^2 \mid \mathcal{F}_t\big] \ \le\ \frac{2\,\overline{L}_{gx}^2}{|B_t^{gx}|}\big(\|x_t - x_{t-1}\|^2 + \|z_t - z_{t-1}\|^2\big) + \frac{2(\gamma_t^x)^2}{|B_t^{gx}|}\,\sigma_{g,x}^2.$$

Using $\|a + \lambda(b-c)\|^2 \le 2\|a\|^2 + 2\lambda^2\|b-c\|^2 \le 2\|a\|^2 + 4\lambda^2(\|b\|^2 + \|c\|^2)$, we obtain

$$\begin{aligned}
\mathbb{E}\big[\|\xi_t^x\|^2 \mid \mathcal{F}_t\big] &\le 2\,\mathbb{E}\big[\|\xi_{t,f}^x\|^2 \mid \mathcal{F}_t\big] + 4\lambda_t^2\left(\mathbb{E}\big[\|\xi_{t,g}^{x,y}\|^2 \mid \mathcal{F}_t\big] + \mathbb{E}\big[\|\xi_{t,g}^{x,z}\|^2 \mid \mathcal{F}_t\big]\right) \\
&\le \frac{2\,\overline{L}_{fx}^2}{|B_t^{fx}|}\big(\|x_t - x_{t-1}\|^2 + \|y_t - y_{t-1}\|^2\big) + \frac{4\,\lambda_t^2\,\overline{L}_{gx}^2}{|B_t^{gx}|}\big(\|x_t - x_{t-1}\|^2 + \|y_t - y_{t-1}\|^2 + \|z_t - z_{t-1}\|^2\big) \\
&\quad + \frac{2(\gamma_t^x)^2}{|B_t^{fx}|}\,\sigma_{f,x}^2 + \frac{4\lambda_t^2(\gamma_t^x)^2}{|B_t^{gx}|}\,\sigma_{g,x}^2.
\end{aligned} \tag{78}$$

Define the inter-round drift at $(x_{t-1}, y_{t-1}, z_{t-1})$ (again allowing λ to vary):

$$\begin{aligned}
\Delta_{t,x}^{L_{\lambda_t},t} := &\Big(\nabla_x f_{t-1}(x_{t-1}, y_{t-1}) + \lambda_{t-1}\big(\nabla_x g_{t-1}(x_{t-1}, y_{t-1}) - \nabla_x g_{t-1}(x_{t-1}, z_{t-1})\big)\Big) \\
&- \Big(\nabla_x f_t(x_{t-1}, y_{t-1}) + \lambda_{t-1}\big(\nabla_x g_t(x_{t-1}, y_{t-1}) - \nabla_x g_t(x_{t-1}, z_{t-1})\big)\Big),
\end{aligned} \tag{79}$$

so that $\epsilon_{t-1}^{x,(t)} = \epsilon_{t-1}^x + \Delta_t^x$, where $\epsilon_{t-1}^x := d_{t-1}^x - \nabla_x f_{t-1}(x_{t-1}, y_{t-1}) - \lambda_{t-1}\big(\nabla_x g_{t-1}(x_{t-1}, y_{t-1}) - \nabla_x g_{t-1}(x_{t-1}, z_{t-1})\big)$. By Young's inequality, for any $\theta_t \in (0,1]$,

$$\|\epsilon_{t-1}^{x,(t)}\|^2 \le (1+\theta_t)\|\epsilon_{t-1}^x\|^2 + \Big(1 + \tfrac{1}{\theta_t}\Big)\left\|\Delta_{t,x}^{L_{\lambda_t},t}\right\|^2. \tag{80}$$

Putting things together. Plug equation 78 and equation 80 into equation 75 with $\theta_t = \gamma_t^x/2$; using $(1-\gamma)^2(1+\gamma/2) \le 1-\gamma/2$, we get

$$\begin{aligned}
\mathbb{E}\big[\|\epsilon_t^x\|^2 \mid \mathcal{F}_t\big] \le &\Big(1 - \tfrac{\gamma_t^x}{2}\Big)\|\epsilon_{t-1}^x\|^2 + \Big(1 - \tfrac{\gamma_t^x}{2}\Big)\Big(1 + \tfrac{2}{\gamma_t^x}\Big)\|\Delta_{t,x}^{L_{\lambda_t},t}\|^2 \\
&+ \left[\frac{2\,\overline{L}_{fx}^2}{|B_t^{fx}|} + \frac{4\,\lambda_t^2\,\overline{L}_{gx}^2}{|B_t^{gx}|}\right]\cdot\big(\|x_t - x_{t-1}\|^2 + \|y_t - y_{t-1}\|^2\big) + \frac{4\,\lambda_t^2\,\overline{L}_{gx}^2}{|B_t^{gx}|}\cdot\|z_t - z_{t-1}\|^2 \\
&+ \frac{2(\gamma_t^x)^2}{|B_t^{fx}|}\,\sigma_{f,x}^2 + \frac{4\lambda_t^2(\gamma_t^x)^2}{|B_t^{gx}|}\,\sigma_{g,x}^2.
\end{aligned} \tag{81}$$

Combining the bounds equation 73 and equation 81 with the already established z-case equation 64 completes the proof of Lemma E.2.

□

Inter-round gradient-drift budgets. For $u \in \{x, y\}$, define the sup-norm one-step drifts

$$\Delta_t^{f,u} \ := \ \big\|\nabla_u f_t(x_{t-1}, y_{t-1}) - \nabla_u f_{t-1}(x_{t-1}, y_{t-1})\big\|, \qquad \Delta_t^{g,u} \ := \ \big\|\nabla_u g_t(x_{t-1}, y_{t-1}) - \nabla_u g_{t-1}(x_{t-1}, y_{t-1})\big\|.$$

We also track the penalty variation $\Delta_t^{\lambda} := |\lambda_t - \lambda_{t-1}|$, and use uniform magnitude envelopes

$$\Gamma_{g,y} \ := \ \sup_{s \le T} \sup_{x,y} \big\|\nabla_y g_s(x, y)\big\|, \qquad \Gamma_{g,x}^{(\text{diff})} \ := \ \sup_{s \le T} \sup_{x,y,z} \big\|\nabla_x g_s(x, y) - \nabla_x g_s(x, z)\big\| \ \le \ 2\,\Gamma_{g,x},$$

with $\Gamma_{g,x} := \sup_{s \le T} \sup_{x,y} \|\nabla_x g_s(x, y)\|$. (If desired, $\Gamma_{g,x}^{(\text{diff})}$ can be further upper bounded via a Lipschitz-in-y constant.)

Lemma E.3 (Bounding the inter-round composites)**.** *Recall*

$$\Delta_{t,y}^{L_{\lambda_t},t} := \big(\nabla_y f_{t-1} + \lambda_{t-1}\nabla_y g_{t-1}\big)(x_{t-1}, y_{t-1}) - \big(\nabla_y f_t + \lambda_t \nabla_y g_t\big)(x_{t-1}, y_{t-1}),$$

$$\begin{aligned}\Delta_{t,x}^{L_{\lambda_t},t} := &\Big(\nabla_x f_{t-1} + \lambda_{t-1}\big(\nabla_x g_{t-1}(\cdot, y) - \nabla_x g_{t-1}(\cdot, z)\big)\Big)(x_{t-1}, y_{t-1}, z_{t-1}) \\ &- \Big(\nabla_x f_t + \lambda_t\big(\nabla_x g_t(\cdot, y) - \nabla_x g_t(\cdot, z)\big)\Big)(x_{t-1}, y_{t-1}, z_{t-1}).\end{aligned}$$

Then

$$\|\Delta_{t,y}^{L_{\lambda_t},t}\|^2 \ \le \ 2\big(\Delta_t^{f,y}\big)^2 + 2\lambda_{t-1}^2\big(\Delta_t^{g,y}\big)^2, \tag{82}$$

$$\|\Delta_{t,x}^{L_{\lambda_t},t}\|^2 \ \le \ 2\big(\Delta_t^{f,x}\big)^2 + 4\lambda_{t-1}^2\big(\Delta_t^{g,x}\big)^2. \tag{83}$$

Proof. For y:

$$\Delta_{t,y}^{L_{\lambda_t},t} = \underbrace{\big(\nabla_y f_{t-1} - \nabla_y f_t\big)(x_{t-1}, y_{t-1})}_{\text{(I)}} + \lambda_{t-1} \underbrace{\big(\nabla_y g_{t-1} - \nabla_y g_t\big)(x_{t-1}, y_{t-1})}_{\text{(II)}}.$$

By definition of the sup-drifts and the envelope, $\|\text{(I)}\| \le \Delta_t^{f,y}$, and $\|\text{(II)}\| \le \Delta_t^{g,y}$. Hence $\|\Delta_t^y\| \le \Delta_t^{f,y} + \lambda_{t-1}\Delta_t^{g,y}$, and $(a+b)^2 \le 2(a^2 + b^2)$ gives the squared bound.

For x:

$$\begin{aligned}\Delta_{t,x}^{L_{\lambda_t},t} = &\underbrace{\big(\nabla_x f_{t-1} - \nabla_x f_t\big)(x_{t-1}, y_{t-1})}_{\text{(I)}} + \lambda_{t-1} \underbrace{\big(\nabla_x g_{t-1} - \nabla_x g_t\big)(x_{t-1}, y_{t-1})}_{\text{(II)}} \\ &- \lambda_{t-1} \underbrace{\big(\nabla_x g_{t-1} - \nabla_x g_t\big)(x_{t-1}, z_{t-1})}_{\text{(III)}}.\end{aligned}$$

Then $\|\text{(I)}\| \le \Delta_t^{f,x}$, $\|\text{(II)}\| \le \Delta_t^{g,x}$, and $\|\text{(III)}\| \le \Delta_t^{g,x}$, so

$$\|\Delta_t^x\| \le \Delta_t^{f,x} + 2\lambda_{t-1}\Delta_t^{g,x},$$

and again $(a+b)^2 \le 2(a^2 + b^2)$ yields equation 83. $\square$

Plugging into the MSE recursions. Applying Lemma E.3 an use the bounds in Equations (73) and (81), we have the following explicit forms:

$$\begin{aligned}\mathbb{E}\big[\|\epsilon_t^y\|^2 \mid \mathcal{F}_t\big] \le &\Big(1 - \tfrac{\gamma_t^y}{2}\Big)\|\epsilon_{t-1}^y\|^2 + 2\Big(1 - \tfrac{\gamma_t^y}{2}\Big)\Big(1 + \tfrac{2}{\gamma_t^y}\Big)\Big[\big(\Delta_t^{f,y}\big)^2 + \lambda_{t-1}^2\big(\Delta_t^{g,y}\big)^2\Big] \\ &+ \left[\frac{2\,\overline{L}_{fy}^2}{|B_t^{fy}|} + \frac{2\,\lambda_t^2\,\overline{L}_{gy}^2}{|B_t^{gy}|}\right] \cdot \big(\|x_t - x_{t-1}\|^2 + \|y_t - y_{t-1}\|^2\big) \\ &+ \frac{2(\gamma_t^y)^2}{|B_t^{fy}|}\sigma_{f,y}^2 + \frac{2\lambda_t^2(\gamma_t^y)^2}{|B_t^{gy}|}\sigma_{g,y}^2,\end{aligned} \tag{84}$$

$$\mathbb{E}\big[\|\epsilon_t^x\|^2 \mid \mathcal{F}_t\big] \le \Big(1-\tfrac{\gamma_t^x}{2}\Big)\|\epsilon_{t-1}^x\|^2 + 2\Big(1-\tfrac{\gamma_t^x}{2}\Big)\Big(1+\tfrac{2}{\gamma_t^x}\Big)\Big[\big(\Delta_t^{f,x}\big)^2 + 2\lambda_{t-1}^2\big(\Delta_t^{g,x}\big)^2\Big]$$
$$+ \left[\frac{2\,\overline{L}_{fx}^2}{|B_t^{fx}|} + \frac{4\,\lambda_t^2\,\overline{L}_{gx}^2}{|B_t^{gx}|}\right]\cdot\big(\|x_t - x_{t-1}\|^2 + \|y_t - y_{t-1}\|^2\big) + \frac{4\,\lambda_t^2\,\overline{L}_{gx}^2}{|B_t^{gx}|}\cdot\|z_t - z_{t-1}\|^2$$
$$+ \frac{2(\gamma_t^x)^2}{|B_t^{fx}|}\sigma_{f,x}^2 + \frac{4\lambda_t^2(\gamma_t^x)^2}{|B_t^{gx}|}\sigma_{g,x}^2. \tag{85}$$

Remarks. If λ_t is constant, the terms with Δ_t^λ vanish and there is no dependence on $\Gamma_{g,y}$ and $\Gamma_{g,x}^{(\mathrm{diff})}$ in equation 84 and equation 85 respectively.

Lemma E.4 (Inner tracking recursion). *Assume $\eta_t^y \le (L_f + \lambda_t L_g)^{-1}$ and $\eta_t^z \le L_g^{-1}$, and the inner updates*

$$y_{t+1} = y_t - \eta_t^y d_t^y, \qquad z_{t+1} = z_t - \eta_t^z d_t^z,$$

with $d_t^y = G_t^y + \epsilon_t^y$ and $d_t^z = G_t^z + \epsilon_t^z$, where

$$G_t^y := \nabla_y f_t(x_t, y_t) + \lambda_t \nabla_y g_t(x_t, y_t), \qquad G_t^z := \nabla_y g_t(x_t, z_t).$$

Let

$$e_t^y := \|y_t - y_{\lambda_t,t}^*(x_t)\|, \quad e_t^z := \|z_t - y_t^*(x_t)\|, \quad \delta_t := (e_t^y)^2 + (e_t^z)^2. \tag{86}$$

Define the one-step minimizer drifts

$$\Gamma_t^{(\lambda)} := \sup_x \|y_{\lambda_{t+1},t+1}^*(x) - y_{\lambda_t,t}^*(x)\|^2, \qquad \Gamma_t := \sup_x \|y_{t+1}^*(x) - y_t^*(x)\|^2.$$

Then there exist per-round constants

$$c_t := \tfrac{1}{2}\min\{\lambda_t\mu\,\eta_t^y,\ \mu\,\eta_t^z\} \in (0,1], \quad K_t^y := \frac{3}{\lambda_t\mu\,\eta_t^y}, \quad K_t^z := \frac{3}{\mu\,\eta_t^z}, \quad C_x := \frac{32L_g^2}{\mu^2}K_t^y + \frac{2L_g^2}{\mu^2}K_t^z \tag{87}$$

such that, conditioning on $\mathcal{F}_t$,

$$\mathbb{E}[\delta_{t+1} \mid \mathcal{F}_t] \le (1-c_t)\,\delta_t \;+\; C_x\,\|\Delta x_t\|^2 \;+\; 2K_t^y\,\Gamma_t^{(\lambda)} + 2K_t^z\,\Gamma_t$$
$$+ \frac{2\eta_t^y}{\lambda_t\mu}\Big(1+\phi_t^y\Big)\mathbb{E}\big[\|\epsilon_t^y\|^2 \mid \mathcal{F}_t\big] + \frac{\eta_t^z}{\mu}\Big(1+\phi_t^z\Big)\mathbb{E}\big[\|\epsilon_t^z\|^2 \mid \mathcal{F}_t\big], \tag{88}$$

where

$$\phi_t^y := \frac{\eta_t^y\lambda_t\mu}{2\,(2-\eta_t^y\lambda_t\mu)}, \qquad \phi_t^z := \frac{\eta_t^z\mu}{2\,(1-\eta_t^z\mu)}.$$

Proof. Fix t.

Step 1: One-step contraction for the current inner problems. Define $F_t(y) := L_{\lambda_t,t}(x_t, y)$ and $G_t(z) := g_t(x_t, z)$, with minimizers $y_t^{\lambda,*} := y_{\lambda_t,t}^*(x_t)$ and $z_t^* := y_t^*(x_t)$. By Assumption 1 and $\lambda_t \ge 2L_f/\mu$, $F_t(\cdot)$ is $(\lambda_t\mu/2)$-strongly convex and $(L_f + \lambda_t L_g)$-smooth in y (Lemma C.2), and $G_t(\cdot)$ is μ-strongly convex and L_g-smooth in z.

Write the deterministic gradient steps and the perturbed steps as

$$\widehat{y}_{t+1} := y_t - \eta_t^y\nabla F_t(y_t), \quad y_{t+1} = \widehat{y}_{t+1} - \eta_t^y\epsilon_t^y, \qquad \widehat{z}_{t+1} := z_t - \eta_t^z\nabla G_t(z_t), \quad z_{t+1} = \widehat{z}_{t+1} - \eta_t^z\epsilon_t^z.$$

For any $\theta > 0$, the weighted Young inequality gives

$$\|y_{t+1} - y_t^{\lambda,*}\|^2 \le (1+\theta)\|\widehat{y}_{t+1} - y_t^{\lambda,*}\|^2 + \Big(1+\tfrac{1}{\theta}\Big)(\eta_t^y)^2\|\epsilon_t^y\|^2,$$

and similarly for z. Using $\eta_t^y \le (L_f + \lambda_t L_g)^{-1}$ and $\eta_t^z \le L_g^{-1}$, the exact steps contract:

$$\|\widehat{y}_{t+1} - y_t^{\lambda,*}\|^2 \le (1 - \tfrac{1}{2}\lambda_t\mu\,\eta_t^y)\|y_t - y_t^{\lambda,*}\|^2, \qquad \|\widehat{z}_{t+1} - z_t^*\|^2 \le (1 - \mu\,\eta_t^z)\|z_t - z_t^*\|^2.$$

Choosing $\theta = \theta_t^y := \frac{\eta_t^y\lambda_t\mu}{2(2-\eta_t^y\lambda_t\mu)}$ and $\theta = \theta_t^z := \frac{\eta_t^z\mu}{2(1-\eta_t^z\mu)}$ gives

$$\mathbb{E}[\|y_{t+1} - y_t^{\lambda,*}\|^2 \mid \mathcal{F}_t] \le (1 - \tfrac{1}{4}\lambda_t\mu\,\eta_t^y)\|y_t - y_t^{\lambda,*}\|^2 + \frac{2\eta_t^y}{\lambda_t\mu}(1+\phi_t^y)\mathbb{E}[\|\epsilon_t^y\|^2 \mid \mathcal{F}_t],$$

$$\mathbb{E}[\|z_{t+1} - z_t^*\|^2 \mid \mathcal{F}_t] \le (1 - \tfrac{1}{2}\mu\,\eta_t^z)\|z_t - z_t^*\|^2 + \frac{\eta_t^z}{\mu}(1+\phi_t^z)\mathbb{E}[\|\epsilon_t^z\|^2 \mid \mathcal{F}_t].$$

Step 2: Shift the targets to $(t{+}1, x_{t+1})$**.** By $(a+b)^2 \le 2a^2 + 2b^2$,

$$\|y_{t+1} - y^*_{\lambda_{t+1},t+1}(x_{t+1})\|^2 \le 2\|y_{t+1} - y_t^{\lambda,*}\|^2 + 2\|y_t^{\lambda,*} - y^*_{\lambda_{t+1},t+1}(x_{t+1})\|^2,$$
$$\|z_{t+1} - y^*_{t+1}(x_{t+1})\|^2 \le 2\|z_{t+1} - z_t^*\|^2 + 2\|z_t^* - y^*_{t+1}(x_{t+1})\|^2.$$

For the drift terms, use Lipschitzness of the minimizer maps in x (Appendix B in Chen et al. [2025a]):

$$\|y^*_{\lambda_{t+1},t+1}(x_{t+1}) - y^*_{\lambda_{t+1},t+1}(x_t)\| \le \frac{4L_g}{\mu}\|\Delta x_t\|, \qquad \|y^*_{t+1}(x_{t+1}) - y^*_{t+1}(x_t)\| \le \frac{L_g}{\mu}\|\Delta x_t\|,$$

and then split the time-variation via $\Gamma_t^{(\lambda)}$ and Γ_t to obtain

$$\|y_t^{\lambda,*} - y^*_{\lambda_{t+1},t+1}(x_{t+1})\|^2 \le \frac{32L_g^2}{\mu^2}\|\Delta x_t\|^2 + 2\Gamma_t^{(\lambda)}, \qquad \|z_t^* - y^*_{t+1}(x_{t+1})\|^2 \le \frac{2L_g^2}{\mu^2}\|\Delta x_t\|^2 + 2\Gamma_t.$$

Step 3: Combine. Adding the y and z bounds, and absorbing constants into c_t, K_t^y, K_t^z, C_x, yields equation 88. □

E.1 PROOF OF THEOREM 5.1

Proof of Theorem 5.1. Throughout, let

$$g_t^* := \nabla L^*_{\lambda_t,t}(x_t), \qquad \Delta x_t := x_{t+1} - x_t = -\eta_t^x d_t^x, \qquad \mathsf{E}_t^x := d_t^x - g_t^*.$$

Also define the population outer direction evaluated at the current inner iterates

$$G_t^x := \nabla_x f_t(x_t, y_t) + \lambda_t\big(\nabla_x g_t(x_t, y_t) - \nabla_x g_t(x_t, z_t)\big), \qquad \epsilon_t^x := d_t^x - G_t^x.$$

Step 1: Outer proxy descent and telescoping. Apply Lemma E.1 to $L^*_{\lambda_t,t}$, take expectations, and sum over $t = 0, \dots, T-1$:

$$\begin{aligned}
&\frac{1}{2}\sum_{t=0}^{T-1} \eta_t^x\, \mathbb{E}\|g_t^*\|^2 \;+\; \frac{1}{4}\sum_{t=0}^{T-1} \frac{1}{\eta_t^x}\, \mathbb{E}\|\Delta x_t\|^2 \\
&\qquad \le L^*_{\lambda_0,0}(x_0) - \mathbb{E}\big[L^*_{\lambda_T,T}(x_T)\big] + \sum_{t=1}^{T} \sup_x \Big| L^*_{\lambda_t,t}(x) - L^*_{\lambda_{t-1},t-1}(x)\Big| + \frac{1}{2}\sum_{t=0}^{T-1} \eta_t^x\, \mathbb{E}\|\mathsf{E}_t^x\|^2 \\
&\qquad \le L^*_{\lambda_0,0}(x_0) - \inf_x L^*_{\lambda_T,T}(x) + V_T^{(\lambda)} + \frac{1}{2}\sum_{t=0}^{T-1} \eta_t^x\, \mathbb{E}\|\mathsf{E}_t^x\|^2.
\end{aligned} \tag{89}$$

Step 2: Bounding the outer inexactness by VR error and inner tracking. Decompose $\mathsf{E}_t^x = (d_t^x - G_t^x) + (G_t^x - g_t^*) = \epsilon_t^x + (G_t^x - g_t^*)$, hence

$$\|\mathsf{E}_t^x\|^2 \le 2\|\epsilon_t^x\|^2 + 2\|G_t^x - g_t^*\|^2.$$

By Lemma C.1, g_t^* equals G_t^x evaluated at $(y_t, z_t) = (y^*_{\lambda_t,t}(x_t),\, y_t^*(x_t))$; therefore, by (x, y)-smoothness,

$$\|G_t^x - g_t^*\| \le (L_f + \lambda_t L_g)\|y_t - y^*_{\lambda_t,t}(x_t)\| + (\lambda_t L_g)\|z_t - y_t^*(x_t)\|.$$

Squaring and using $\delta_t = \|y_t - y^*_{\lambda_t,t}(x_t)\|^2 + \|z_t - y_t^*(x_t)\|^2$ yields

$$\|\mathsf{E}_t^x\|^2 \le 2\|\epsilon_t^x\|^2 + C_{e,t}\,\delta_t, \qquad C_{e,t} := 4\Big((L_f + \lambda_t L_g)^2 + (\lambda_t L_g)^2\Big). \tag{90}$$

Step 3: Summing the inner recursion and absorbing coupling terms. From Lemma E.4 and equation 88, taking expectations and summing over $t = 0, \dots, T-1$ gives

$$\sum_{t=0}^{T-1} c_t\, \mathbb{E}[\delta_t] \le \mathbb{E}[\delta_0] \;+\; \sum_{t=0}^{T-1} C_{x,t}\, \mathbb{E}\|\Delta x_t\|^2 \;+\; 2\sum_{t=0}^{T-1}\Big(K_t^y\, \Gamma_t^{(\lambda)} + K_t^z\, \Gamma_t\Big)$$
$$+ \sum_{t=0}^{T-1} A_{y,t}\, \mathbb{E}\|\epsilon_t^y\|^2 \;+\; \sum_{t=0}^{T-1} A_{z,t}\, \mathbb{E}\|\epsilon_t^z\|^2, \tag{91}$$

where $A_{y,t} := \frac{2\eta_t^y}{\lambda_t \mu}(1+\phi_t^y)$, $A_{z,t} := \frac{\eta_t^z}{\mu}(1+\phi_t^z)$, and we write $C_{x,t}, K_t^y, K_t^z$ for the constants in Lemma E.4.

Combine equation 89, equation 90, and equation 91. If, for all t,

$$\eta_t^x\, C_{x,t} \le \frac{1}{8}, \qquad \eta_t^x\, C_{e,t} \le \frac{c_t}{2}, \tag{92}$$

then the $\sum_t C_{x,t}\mathbb{E}\|\Delta x_t\|^2$ term is absorbed by the movement mass $\sum_t \frac{1}{\eta_t^x}\mathbb{E}\|\Delta x_t\|^2$ in equation 89, and the $\sum_t \eta_t^x C_{e,t}\mathbb{E}[\delta_t]$ term is absorbed by $\sum_t c_t \mathbb{E}[\delta_t]$ in equation 91. Dropping the remaining nonnegative terms on the left-hand side, we obtain

$$\sum_{t=0}^{T-1} \eta_t^x\, \mathbb{E}\|g_t^*\|^2 \le C\Big(L^*_{\lambda_0,0}(x_0) - \inf_x L^*_{\lambda_T,T}(x) + V_T^{(\lambda)} + \mathbb{E}[\delta_0] + \sum_{t=0}^{T-1}\big(\Gamma_t^{(\lambda)} + \Gamma_t\big)\Big) + C\sum_{t=0}^{T-1} \eta_t^x\, \mathbb{E}\|\epsilon_t^x\|^2 + C\sum_{t=0}^{T-1}\Big(A_{y,t}\mathbb{E}\|\epsilon_t^y\|^2 + A_{z,t}\mathbb{E}\|\epsilon_t^z\|^2 \tag{93}$$

for a constant C polynomial in (μ^{-1}, L_f, L_g).

Step 4: From proxy gradients to $\nabla\phi_t$. By Lemma C.1, $\|\nabla\phi_t(x_t)\| \le \|g_t^*\| + D_1/\lambda_t$, hence $\|\nabla\phi_t(x_t)\|^2 \le 2\|g_t^*\|^2 + 2D_1^2/\lambda_t^2$. Multiply by η_t^x, sum, divide by $S_x = \sum_{t=0}^{T-1}\eta_t^x$, and apply equation 93 to obtain equation 13.

Finally, Lemma E.2 and Lemma E.3 provide explicit upper bounds on the error sums $\sum_t \eta_t^x \mathbb{E}\|\epsilon_t^x\|^2$, $\sum_t A_{y,t}\mathbb{E}\|\epsilon_t^y\|^2$, and $\sum_t A_{z,t}\mathbb{E}\|\epsilon_t^z\|^2$ in terms of the drift budgets Equations (9) and (10) and the oracle variances. □